# Bayesian Deep Learning Reveals the Role of Biotic and Abiotic Metrics in Predicting Estuarine Nekton Populations

Liying Li[1], Michael J. Schram[2], Hallie C. Repeta[3], Cameron Ainsworth[3], Kerry Flaherty-Walia[4]
[1]National Center for Ecological Analysis and Synthesis, University of California, Santa Barbara
[2]Florida Fish and Wildlife Conservation Commission, Fish and Wildlife Research Institute, 100 8th Ave SE, St. Petersburg, FL 33701
[3]College of Marine Science, University of South Florida. 140 7th Ave. S., St. Petersburg, FL. 33701
[4]Tampa Bay Estuary Program, 263 13th Ave. S., Suite 350, St. Petersburg, FL 33701

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## Abstract

Understanding why species populations decline and how to assist population recovery has become one of the central challenges in ecosystem management and conservation. Existing predictive models are often limited by spatial and temporal autocorrelation, data scarcity for rare species, complex biotic interactions, and confounding environmental and management drivers. We developed a Bayesian Graph Neural Network with nested Long Short-Term Memory architecture (Bayesian GNN–nLSTM) that integrates graph neural networks, nested temporal modeling, Bayesian transfer learning, and interpretable predictor attribution within a unified framework. Applied to 27 years of fisheries-independent monitoring data from Florida estuaries, the framework accurately predicted fish and macroinvertebrate (nekton) abundance across multiple species and life stages. Biotic community composition was the strongest predictor of model accuracy, indicating that community structure captures substantial ecological information beyond the measured environmental variables. In contrast, habitat and management variables, including shoreline characteristics, submerged aquatic vegetation, and restoration activities, exerted stronger influences on predicted population abundance, suggesting that these factors primarily determine the abundance ecosystems can support rather than fine-scale spatial distributions. Lower predictive performance for recruit stages further indicates that stochastic early-life processes remain difficult to capture using environmental predictors alone.

Keywords: Population ecology, Fishery management, Restoration, Species distribution modeling, Bayesian deep learning, Graph neural networks

## 1. Introduction

Sustaining nekton populations and the ecosystem services they provide has become increasingly challenging as climate change, habitat degradation, coastal development, pollution, and fishing pressure simultaneously reshape aquatic ecosystems. Rising temperatures alter thermal niches and drive shifts in species distributions and phenology (Pörtner & Farrell, 2008; Cheung et al., 2010; Pinsky et al., 2013; Poloczanska et al., 2013), while increasing hypoxia, sea-level rise, altered freshwater inflows, and habitat loss further modify estuarine environments that support nekton recruitment and productivity (Diaz & Rosenberg, 2008; Waycott et al., 2009;

Gillanders et al., 2011; Kirwan & Megonigal, 2013). In response, substantial restoration efforts—including seagrass restoration, shoreline stabilization, artificial reefs, and watershed-scale nutrient reduction—have been implemented to improve habitat quality and enhance fisheries productivity (Lotze et al., 2006; Orth et al., 2006; Bayraktarov et al., 2016). Because these environmental, biological, and management processes operate simultaneously, effective fisheries management requires understanding not only of how environmental change influences species distributions, but also how habitat restoration and biological interactions jointly determine population responses.

Species distribution models (SDMs) have become essential tools for predicting nekton abundance and distributions. In the Gulf of America, hereafter referred to as the Gulf, successive generations of SDMs have progressively improved predictions by accounting for zero inflation, ontogenetic habitat shifts, and spatial autocorrelation using fisheries-independent monitoring data (Drexler & Ainsworth, 2013; Grüss et al., 2014, 2018, 2019; Rubec et al., 2016; Ainsworth et al., 2025). Such efforts have substantially advanced habitat-based fisheries modeling. However, most existing SDMs evaluate abiotic conditions, habitat characteristics, restoration activities, and species interactions separately or rely on relatively simple statistical relationships that cannot adequately represent the nonlinear and spatiotemporally dynamic processes governing nekton populations (Levin, 1992; Planque et al., 2011; Bosso et al., 2024). Consequently, they have limited ability to distinguish whether changes in abundance arise from climate variability, habitat degradation, biotic interactions, restoration actions, or their confounded effects. Moreover, rare species remain difficult to model because limited observations constrain statistical inference (Li et al., 2026).

Recent advances in artificial intelligence offer new opportunities to overcome these limitations. Graph neural networks (GNNs) explicitly model spatial connectivity and ecological relationships (Wu et al., 2021), whereas recurrent neural networks (RNNs), particularly long short-term memory (LSTM) architectures, learn temporal dependencies from sequential observations and have demonstrated strong performance for spatiotemporal forecasting (Reichstein et al., 2019; Fan et al., 2021). In parallel, Bayesian transfer-learning approaches such as common-to-rare transfer learning improve inference for data-limited species by borrowing statistical strength from ecologically related species (Ovaskainen et al., 2025). Despite these advances, few ecological models integrate spatial structure, temporal dynamics, Bayesian information sharing, and interpretable ecological attribution within a unified framework. As a result, the relative importance of climatic variability, habitat structure, community composition, and restoration actions remains poorly resolved, particularly in dynamic estuarine ecosystems where these drivers co-vary across space and time.

To address these challenges, we developed a Bayesian Graph Neural Network with a nested Long Short-Term Memory architecture (Bayesian GNN-nLSTM) to model the dynamics of fish and selected macroinvertebrate populations while disentangling multiple interacting ecological drivers. Our framework addresses four major limitations of existing species distribution models. First, it explicitly accounts for spatial and temporal autocorrelation by combining graph neural networks with a nested LSTM architecture and a shifting-window training strategy that augments long-term monitoring data while simultaneously learning seasonal (inner LSTM) and interannual (outer LSTM) ecological dynamics. Second, it addresses data scarcity and uneven predictive performance between common and rare species through Bayesian transfer learning, enabling information sharing across ecologically related taxa while providing uncertainty-aware predictions.

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Third, it jointly integrates environmental, habitat, community, and restoration predictors within a unified framework, enabling the relative contributions of abiotic, biotic, and management drivers to be evaluated while reducing confounding influences among these interacting processes. Finally, it quantifies the magnitude and direction of predictor effects through a group-wise permutation attribution framework, enabling mechanistic interpretation of the drivers of abundance patterns and supporting ecosystem-based management and restoration planning. Although developed using estuarine fisheries-independent monitoring data, the framework is general and readily transferable to other species, ecosystems, geographic regions, and ecological applications where long-term, heterogeneous monitoring data are available. The Bayesian GNN-nLSTM supports ecosystem-based fisheries management, restoration prioritization, and climate adaptation. Specifically, the framework enabled biodiversity monitoring programs facing similar challenges of sparse observations, complex species interactions, rapidly changing environmental conditions, and evolving conservation actions.

# 2. Method and Materials

## 2.1. Study area and Fisheries-Independent Monitoring dataset

The Bayesian GNN-nLSTM model integrated four broad categories of predictors: (1) biotic, (2) environmental and structural, (3) survey conditions and missing data indicator, and (4) restoration (Table 1). Except for restoration, all predictors were derived from long-term estuarine nekton data (1998-2024) collected by the Florida Fish and Wildlife Conservation Commission's Fisheries-Independent Monitoring (FIM) program and included both direct sampling measurements and derived predictors from those observations. Additional details about the FIM program may be found in McMichael (1991) and FWC-FWRI (2023). However, the sampling protocol and methods necessary to interpret model development are briefly summarized below. Restoration predictors were compiled from the Florida Estuarine Restoration Database, a comprehensive inventory of habitat restoration and estuarine resource improvement projects implemented throughout Florida estuaries and their contributing watersheds.

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**Table 1.** Predictor list used by Bayesian GNN-nLSTM. Predictors included raw and calculated metrics from the FIM or restoration project datasets (1998-2024).

| Group | Number of predictors | Group description |
|---|---|---|
| **Biotic predictors** | | |
| Prey nekton community | 49 | Log-transformed catch counts of small-bodied individuals (21.3-m seine) — shoreline and shallow flats co-occurring assemblage |
| Non-prey nekton community | 45 | Log-transformed catch counts of large-bodied individuals (183-m seine) — shoreline co-occurring assemblage |
| Prior species mean ($\bar{Y}$) | 12 | Station-level historical mean log-transformed catch counts per target species — temporal baseline |
| **Environmental and structural predictors** | | |
| Environmental | 6 | Water temperature, salinity, dissolved oxygen, Secchi depth, Secchi-on-bottom flag, tidal state |
| Bycatch assemblage | 14 | Non-fish material caught: bivalves, seagrass, algae, debris, SAV, crabs, substrate, jellyfish, invertebrates, biotic structure, |

| | | |
|---|---|---|
| | | and miscellaneous categories — proxy indicators of habitat condition |
| Bottom cover type | 7 | Total bottom vegetation cover; substrate fractions (mud, sand, shell/hard, unknown); dominant bottom substrate category; vegetation presence flag |
| Submerged aquatic vegetation | 7 | Vegetation cover by type (SAV, algae, bare, *Halodule, Thalassia, Ruppia*); dominant vegetation category |
| Shoreline | 7 | Shoreline type scores 0–10 (natural, overhanging, structural, emergent); dominant shoreline category; distance to nearest shore; distance to mean high-tide mark |
| **Survey conditions and missingness** | | |
| Sampling efforts | 3 | Cloud cover, start depth, haul count |
| Missing data indicator | 6 | Binary missingness flags for features with >5% missing data |
| **Restoration predictors** | | |
| Restoration | 8 | Number of projects, cumulative acres restored, primary target habitat type, secondary target habitat type, restoration technique type (×2), number of target habitat types, number of restoration techniques |
| **Total** | **164** | |

## 2.2. Prediction targets and predictors

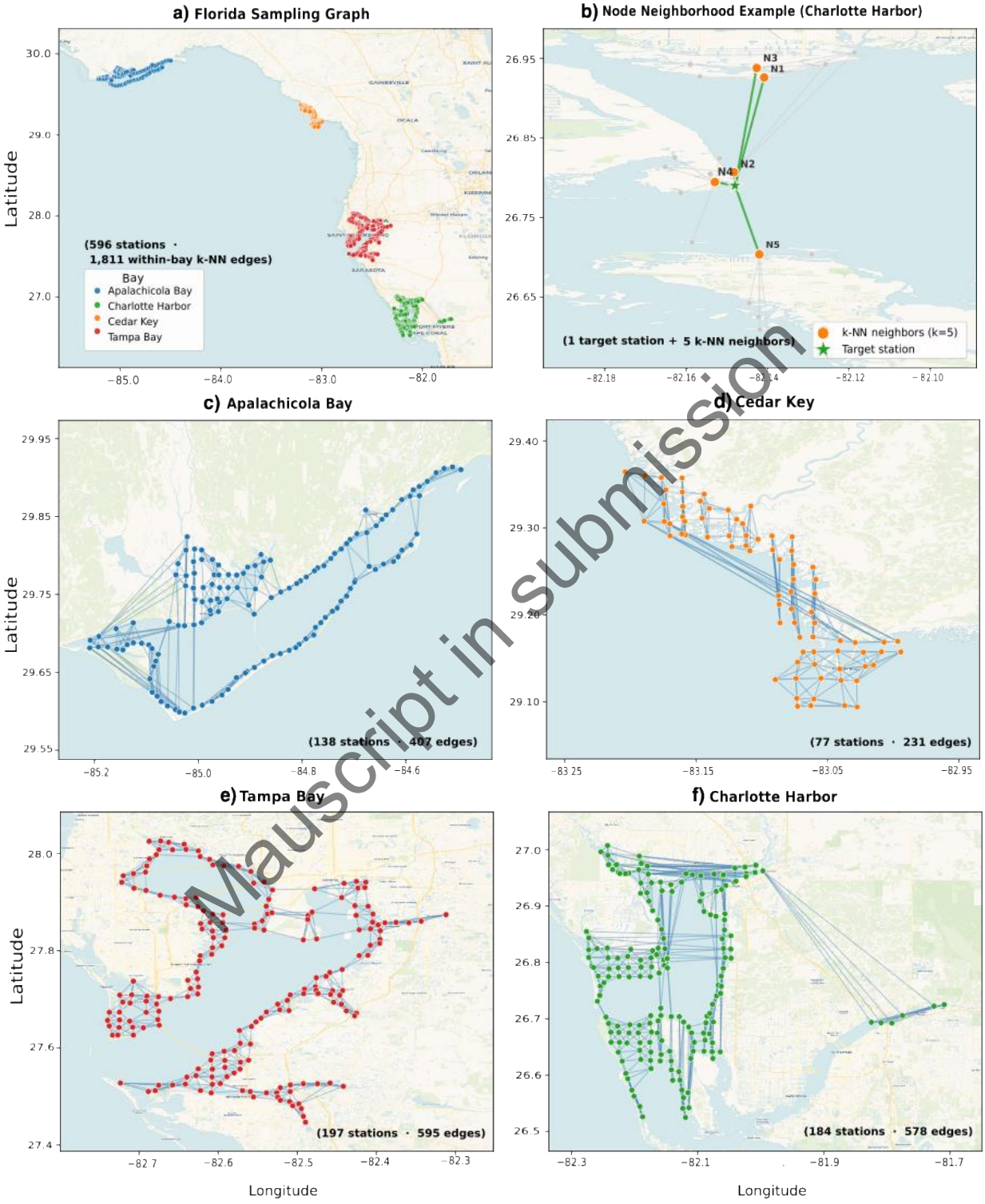
### 2.2.1. Nekton target species and life stages

The FIM program follows a stratified-random sampling design, with monthly effort allocated across spatial zones defined by geographic and hydrological characteristics within four major estuaries located along the Gulf coast of Florida: Apalachicola Bay, Cedar Key, Tampa Bay, and Charlotte Harbor (Figure 1). Multiple gear types are used to sample nekton communities within each estuary, as described in detail below. For each gear type within each estuary, sampling sites are further stratified by habitat type and selected randomly without replacement each month, with effort proportional to the number of available sites in each stratum. All fish and macroinvertebrates collected are identified to the lowest possible taxonomic level (usually species) and enumerated, and as many as 40 randomly selected individuals per taxon within each sample are measured to the nearest millimeter standard length (SL).

Juvenile and resident small-bodied (typically <100 mm SL) fish and macroinvertebrates are sampled using a 21.3-m × 1.8-m center-bag haul seine with 3.2-mm mesh. In shallow habitats (≤1.8 m depth), the seine is deployed over 9.1 m with a width of 15.5 m, sampling approximately 140 m² (21.3-m bay seine). Sampling targets shoreline habitats (e.g., emergent vegetation, mangrove fringes, seawalls, beaches) and offshore flats ≥5 m from shore. Offshore sampling (except at Cedar Key) is stratified by ≥25% cover of submerged aquatic vegetation (e.g., seagrasses, SAV). In riverine systems with steep shorelines, the seine is deployed from a boat in a semi-elliptical configuration and retrieved onto shore, sampling approximately 68 m² (21.3 m river seine).

Subadult and adult fish and macroinvertebrates (typically ≥100 mm SL) are collected using a 183-m × 2.5-m center-bag haul seine with 38-mm mesh (183-m haul seine). The net is deployed by boat in a rectangular configuration along the shoreline, sampling approximately 4,120 m² (40 m × 103 m). In Tampa Bay and Charlotte Harbor, sampling is stratified by the presence of overhanging vegetation (≥10% shoreline cover).

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**Figure 1.** Study areas and spatial representation of the constructed Graph Neural Networks. FIM Gulf Coast estuaries (a); example of a node neighborhood for a single station (b); Graph Neural Network for Apalachicola Bay (c), Cedar Key (d), Tampa Bay (e), and Charlotte Harbor (f).

To limit the influence of extremely sparse taxa on model fitting, species occurring in fewer than 5% of samples (McCune et al., 2002) within each gear-estuary combination (e.g., 21.3-m bay seine in Charlotte Harbor) were excluded. This filtering approach was applied separately for each gear-estuary combination to account for differences in regional taxon pools across the broad latitudinal extent of FIM sampling along Florida's Gulf coast. From the remaining taxa, seven ecologically and economically important taxa were selected as prediction targets, with each life stage (recruit, subadult, or adult) treated as an independent response variable (Table 2). Life stages were defined using species-specific size ranges that correspond to ontogenetic stages relevant to each gear type. We used published information on each species' life history and gear-based length-frequency limitations to inform these size ranges (Switzer et al., 2012; Flaherty et al., 2014; FWC-FWRI, 2023). A log1p transformation was applied to all abundance data to compress the range of observed abundances and reduce the influence of extreme counts on model fitting.

### 2.2.2. Biotic predictors

To represent the expected local abundance of each focal species, we calculated prior species mean ($\bar{Y}$), defined as the mean log1p-transformed abundance of that species at the corresponding station–month across the training years. This feature provides a location- and season-specific baseline that captures persistent spatial and seasonal differences in species abundance. The model can therefore learn departures from typical community conditions associated with environmental variability, habitat conditions, and restoration rather than having to reconstruct the underlying spatial and seasonal baseline from environmental covariates alone. Prior means were calculated using training data only to prevent information leakage into validation and test years.

Community composition was represented separately using catches from the two seine gears employed by the FIM program, which characterize different components of the estuarine fish assemblage. The prey nekton community was derived from log1p-transformed catch counts collected with the 21.3-m seine. These samples primarily characterize fishes and invertebrates occupying shallow shoreline and nearshore habitats and provide information on the local assemblage co-occurring with the focal species, including potential prey and other small-bodied taxa. Rather than assuming that all captured taxa were direct prey, these variables were treated as indicators of the broader small-bodied community potentially associated with local food availability and habitat conditions.

Non-prey nekton community composition was derived from log1p-transformed catch counts collected with the 183-m seine. The larger seine samples a broader and generally more mobile component of the estuarine fish assemblage and therefore provides complementary information on species co-occurring with the focal species at larger spatial and body-size scales. These variables were interpreted as indicators of the surrounding fish community rather than as a predefined trophic group, allowing the model to learn associations with potential competitors, predators, prey, and species responding to similar environmental conditions.

Together, the prior species mean and observed community features provided complementary information. The prior mean represented the long-term expectation for a species at a given location and calendar month, whereas the two seine-derived assemblages represented the community conditions observed during a particular sampling period.

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**Table 2**. List of predicted species where specific life stages were evaluated due to their economic significance.

| Common name | Scientific name | Life stage | Gear | Lower size limit (standard length; mm) | Upper size limit (standard length; mm) |
|---|---|---|---|---|---|
| Red Drum | *Sciaenops ocellatus* | Recruit | 21.3-m bay seine | 0 | 40 |
| Red Drum | *Sciaenops ocellatus* | Recruit | 21.3-m river seine | 0 | 40 |
| Red Drum | *Sciaenops ocellatus* | Subadult | 183-m haul seine | 250 | 700 |
| Spotted Seatrout | *Cynoscion nebulosus* | Recruit | 21.3-m bay seine | 0 | 100 |
| Spotted Seatrout | *Cynoscion nebulosus* | Recruit | 21.3-m river seine | 0 | 100 |
| Spotted Seatrout | *Cynoscion nebulosus* | Adult | 183-m haul seine | 200 | Inf |
| Gray Snapper | *Lutjanus griseus* | Recruit | 21.3-m bay seine | 0 | 50 |
| Gray Snapper | *Lutjanus griseus* | Recruit | 21.3-m river seine | 0 | 50 |
| Gray Snapper | *Lutjanus griseus* | Subadult | 183-m haul seine | 100 | 250 |
| Gag | *Mycteroperca microlepis* | Subadult | 183-m haul seine | 100 | 300 |
| Common Snook | *Centropomus undecimalis* | Subadult | 21.3-m river seine | 100 | 299 |
| Common Snook | *Centropomus undecimalis* | Adult | 183-m haul seine | 300 | Inf |
| Sheepshead | *Archosargus probatocephalus* | Recruit | 21.3-m bay seine | 0 | 40 |
| Sheepshead | *Archosargus probatocephalus* | Recruit | 21.3-m river seine | 0 | 40 |
| Sheepshead | *Archosargus probatocephalus* | Adult | 183-m haul seine | 150 | Inf |
| Blue Crab | *Callinectes sapidus* | Recruit | 21.3-m bay seine | 0 | 80 |
| Blue Crab | *Callinectes sapidus* | Recruit | 21.3-m river seine | 0 | 80 |

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### 2.2.3. Environmental and structural predictors

During each FIM sampling event, environmental variables, including temperature (°C), salinity (psu), and dissolved oxygen (mg $L^{-1}$), are measured at the surface and at 1-m depth intervals for each sample. Habitat characteristics, including shoreline type, bottom substrate, and SAV cover, are recorded at the time of sampling for each net deployment. In addition, composition of bycatch, defined as non-fish materials inadvertently sampled by the seine, acts as an indirect but highly informative indicator of local habitat conditions and ecosystem state. It captures real-time signals of vegetation presence, structural complexity, and environmental stressors that may not be fully represented by static habitat measurements (Stoner & Ottmar, 2004). Bycatch volume and composition (e.g., free-floating algae or seagrass, debris, mud) are also recorded, as these factors may influence nekton composition and abundance within the catch (Rydene & Matheson, 2003).

Our preliminary analyses indicated that differences between surface and bottom measurements of temperature, salinity, and dissolved oxygen within samples were small relative to spatial and temporal variation. Accordingly, these measurements were averaged across the water column for each sample. Habitat variables, including bottom type, vegetation, and shoreline type, were summarized at the sample level to characterize local environmental conditions. In addition, bycatch volume and composition are recorded and serve as indirect but highly informative indicators of local habitat conditions and ecosystem state. Bycatch captures real-time signals of vegetation presence, structural complexity, and environmental stressors that may not be fully represented by static habitat measurements (Stoner & Ottmar, 2004). These factors may influence nekton composition and abundance within the catch (Rydene & Matheson, 2003). Bycatch components include, but are not limited to, macroalgae, seagrasses, structure-forming biota (e.g., corals, sponges), noncommercial invertebrates (e.g., gastropods, tunicates), and debris (e.g., crab traps, woody debris).

Habitat and shoreline features describe fine-scale and landscape-level structure, influencing refuge, foraging opportunities, and connectivity, while survey condition variables correct for observation bias to ensure that predictions reflect true abundance rather than differences in sampling.

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### 2.2.4. Sampling efforts and missing value indicators

Cloud cover, sampling depth, and haul count predictors describe survey conditions rather than habitat characteristics. Cloud cover and sampling depth can influence species behavior and habitat use, thereby affecting encounter rates during sampling, whereas haul count (i.e., the number of sampling events associated with each GNN station) reflects variation in sampling intensity, with more hauls increasing the probability of encountering a species. Haul count is the number of individual sampling events that were aggregated into a station in the GNN. Although site selection is randomized and occurs without replacement, it is still possible for samples from different gear-strata combinations to occur within the same Bay-Zone-Grid cell each month. Therefore, haul count captures how many of those individual net deployments contributed to species counts and environmental measurements within a station. Including these variables allowed the model to account for observation bias while separating sampling effects from ecological drivers of abundance.

Missing values of the FIM dataset were imputed using a hierarchical median approach. Missing observations were first replaced with the median value within the same Bay × Zone × Month grouping; if no observations were available, the global median was used. Variables with

5% or greater missingness were treated as potentially informative, and an additional binary indicator variable (1 = missing, 0 = observed) was included to allow the model to capture whether missingness itself reflected ecological or sampling conditions (e.g., variables recorded only under specific environmental states). For variables with less than 5% missing values, missingness was assumed to be approximately random and no indicator variable was included.

### 2.2.5. Restoration predictors

The four Gulf Coast estuaries sampled using FIM methods have also been the focus of substantial restoration investments aimed at improving habitat quality, enhancing fisheries productivity, and increasing ecosystem resilience. Restoration predictors provide a longer-term context of habitat quality, reflecting cumulative human investment and shifting baseline expectations of biodiversity.

We incorporated Florida estuarine restoration data to explicitly evaluate the influence of restoration activities on estuarine nekton communities. The Florida estuarine restoration database integrates project records from multiple federal, state, regional, and non-governmental programs into a standardized geospatial database. Project records include information describing the spatial and temporal characteristics of each restoration project (e.g., coordinates and completion dates), the ecological focus (e.g., target habitat type(s) and restoration techniques), measures of restoration extent (e.g., acres or miles restored), implementation and funding details (e.g., lead organization, primary funding source, associated legislation, and project costs), and, if available, socioeconomic outcomes (e.g., the number of jobs created). Selected variables from restoration project records were incorporated as spatially explicit predictors in the modeling framework to quantify their relative importance alongside environmental and habitat variables and to evaluate how restoration influences nekton abundance, distribution, and recruitment.

We linked restoration projects to the FIM sampling data using a spatial–temporal matching approach. Each FIM sampling location is a spatial point with a 1,500 m-diameter buffer around it, and restoration project data were cleaned, standardized (Z-scores), and spatially joined to retain only projects within this buffer for each sampling location. For each FIM sampling event, restoration projects within the associated buffer were filtered using a five-year temporal window relative to the sampling year, retaining only projects completed within the five years prior to, and including, that year. This captures recent restoration activity while accounting for potential lagged ecological responses. The retained projects were then summarized into FIM sampling event-level restoration predictors.

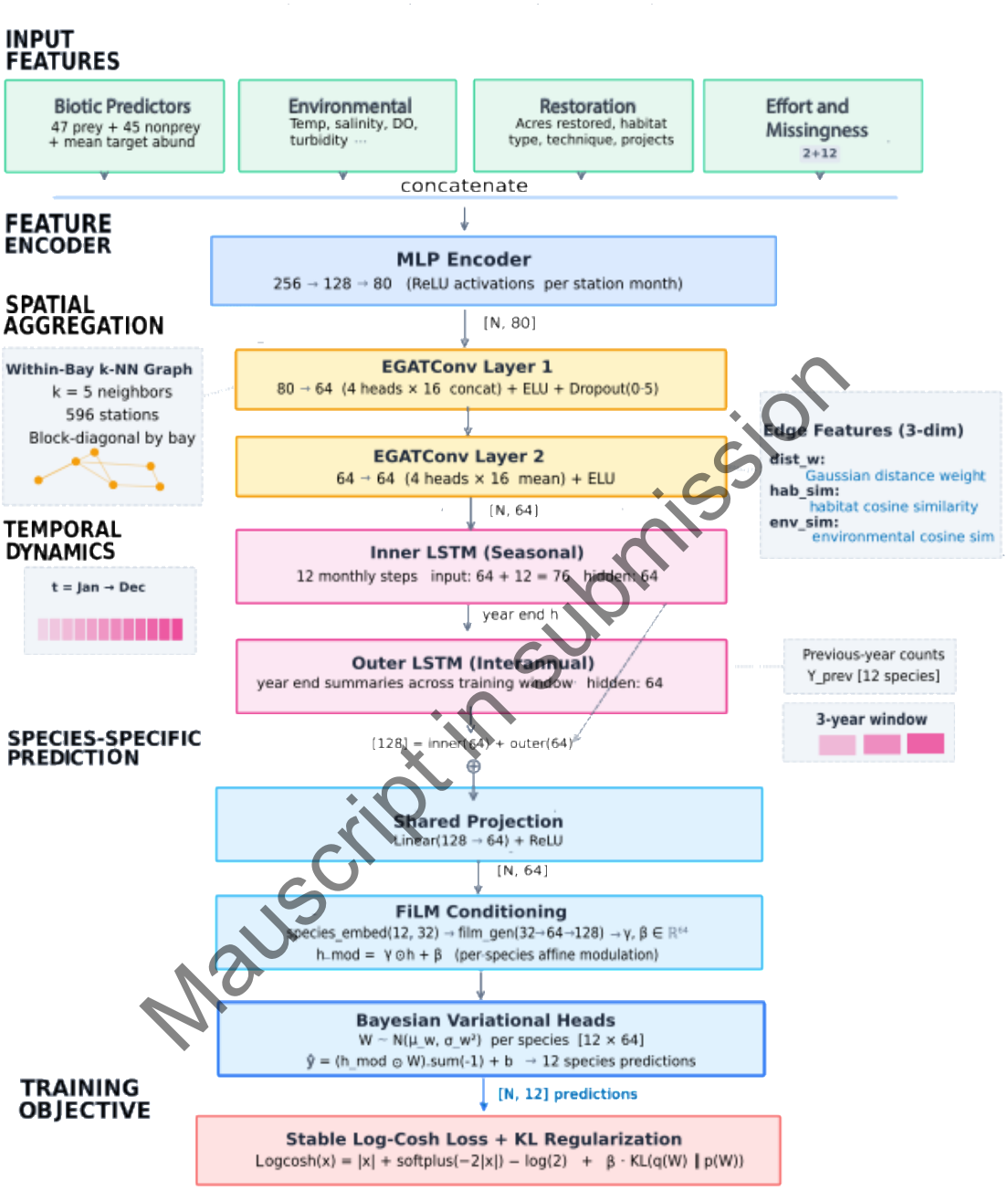
The predictors describe cumulative restoration efforts within each given buffer, including the number of projects and the total area restored (acres restored). Acres restored were summed across projects within a buffer. Primary and secondary targeted habitat types, and first and second restoration techniques, were each determined as the most frequently reported category (mode) across all retained restoration projects within the 1,500-m buffer and 5-year window. When a station was associated with a single project, both fields reflected that project's reported habitat types. We also included the total number of unique habitat types targeted and restoration techniques by averaging the targeted habitat number and restoration technique number across projects.

## 2.3. GNN-nLSTM model architecture

GNN-nLSTM operates on a four-dimensional input tensor of shape [N×T×M×F], corresponding to stations, years, months, and features, respectively (Figure 2). The architecture

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consists of three sequential stages—feature encoding, spatial aggregation, and temporal modeling—followed by species-specific output heads.




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**Figure 2**. Bayesian Graph Neural Networks and Nested Long-Term Short-Term Models combined framework overview

A station was defined as a unique FIM seine sampling location identified by a fixed latitude–longitude coordinate within one of four estuarine systems: Apalachicola Bay, Charlotte Harbor, Cedar Key, and Tampa Bay (Figure 1). Because fish communities exhibit strong seasonal variation, observations were represented at the station–month level rather than using a single static node for each sampling location. Specifically, each physical station was represented

by 12 nodes corresponding to the 12 calendar months, such that each node represented a unique combination of sampling location and calendar month. For (S) stations, the resulting graph therefore contained up to (12S) spatiotemporal nodes (e.g., 7,152 nodes for 596 stations).

At each station–year–month combination, the concatenated feature vector (including environmental variables, community composition, and prior species means ($\bar{Y}$) is projected into a shared latent space (feature compression) using a multi-layer perceptron. The encoder takes 164 environmental and biological variables and transforms them into 80 synthetic variables used by the GNN. The encoder has fully connected layers with hidden dimensions 256 and 128, followed by an 80-dimensional output projection (164 → 256 → 128 → 80), with rectified linear unit (ReLU) activations applied between layers. Each of the four layers learns increasingly abstract combinations of the original variables by simultaneously optimizing weights used to connect neurons to successive layers and thus minimizing a loss function comparing predicted and observed fish abundance. This layer structure provides sufficient capacity while reducing the risk of overfitting. This step performs nonlinear compression of heterogeneous predictors into a unified representation that serves as input to the spatial aggregation stage by the GNN.

Spatial dependencies among stations are modeled using a two-layer Graph Attention Network (GAT) with edge-aware attention. The first GAT layer aggregates neighbor information using multi-head concatenation, while the second layer averages the outputs across heads. Both layers maintain a fixed 64-dimensional node representation. This means that each node's representation is influenced by neighboring nodes, with the contribution of each neighbor determined by its learned attention coefficients. Stations are connected via a k-nearest neighbor graph (K=5), constrained within each estuary to preserve hydrological realism and prevent connections across bay boundaries. Attention coefficients are computed by conditioning on both node and edge features, allowing the model to learn context-dependent spatial relationships. This formulation enables the model to weigh neighbors based on both ecological similarity and physical connectivity.

Specifically, for each attention head, a weighted sum of neighbor embeddings is computed, and weights α are learned for each edge; all five heads were concatenated to form the node's updated representation:

$$H_v^1 = \|_{k=1}^{K} \sigma\left( \Sigma_{u \in N(v)} \alpha_{vu}^k W^k h_u \right) \quad (1)$$

where $H_v^1$ is the output embedding of node v after the 1st attention layer. $\|_{k=1}^{K}$ is the concatenation of K=5 attention heads (each head learns a different aspect of the neighborhood). σ is a nonlinear activation function (Exponential Linear Unit, ELU). $\boldsymbol{\Sigma_{u \in N(v)}}$ is the sum over all neighbor nodes $u$ in the neighborhood of node $v$. $N(v)$ is the set of neighbors of node v (determined by the adjacency matrix). $W^k$ denotes the learnable weight matrix for head k out of all four heads, which projects the input embedding into that head's representation space. $h_u$ is the input feature embedding of neighbor node $u$ (output of the encoder). $\alpha_{vu}^k$ is the learned attention coefficient from head $k$ between node $v$ and neighbor $u$; the main purpose of GAT is edge-aware attention. $\alpha_{vu}^k$ controls how much $u$'s information influences $v$ (these are softmax-normalized across all neighbors). Each edge is associated with a three-dimensional feature vector encoding inverse geographic distance, habitat similarity, and environmental similarity between station pairs. Habitat similarity between station pairs was quantified as the cosine similarity of per-station median habitat and shoreline feature vectors (21 feature predictors), computed across all available sampling years. Environmental similarity was computed analogously using median

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profiles of the four most stable environmental variables: water temperature, salinity, dissolved oxygen, and Secchi depth. For both measures, feature vectors were unit-normalized prior to computing the dot product, yielding similarity values in [−1,1]. The third edge attribute, geographic distance weight, was computed as a Gaussian decay function

Temporal dynamics are modeled using a hierarchical (nested) two-level LSTM. The inner LSTM processes the 12 monthly timesteps within each year, capturing seasonal dynamics and conditioning on lagged same-month observations from the previous year to model seasonal persistence. These inner LSTMs can learn and account for shifts in fish recruitment timing over the 30-year study period. Monthly dynamics reflect shifting between ecologically important periods (e.g., peak recruitment) and off-season observations.

The outer LSTM operates on a sequence of yearly summaries produced by the inner LSTM, integrating information across time to capture interannual trends and long-term population dynamics. The outer LSTM processes a sequence of 10 such windows (Figure 2), providing an effective temporal context of approximately 30 years. This hierarchical design separates short-term seasonal variability from longer-term population trajectories and enables the model to capture lagged ecological responses. Temporal inputs are structured using rolling windows.

## 2.4. Rolling-window temporal representation

To represent long-term ecological dynamics while reducing the influence of highly variable annual observations, we adopted a nested rolling-window temporal representation. Rather than using individual years as independent inputs, the model constructs a sequence of ten overlapping three-year windows for each prediction year (Figure 3). Each window summarizes ecological conditions over three consecutive years, and successive windows advance by one year, producing a continuous temporal trajectory spanning approximately twelve calendar years.

For example, when predicting 2024, the rolling window consists of ten overlapping three-year periods that progress forward one year at a time, extending from 2013–2015 to 2022–2024 for the 2024 prediction. Thus, each prediction is informed by a temporally overlapping history spanning approximately 12 years while preserving gradual year-to-year transitions. Within each window, predictor variables and observed abundances are averaged across all available years. Years with missing observations are excluded from the averaging procedure rather than treated as zero, preventing artificial reductions in abundance estimates for sparsely sampled stations.

The resulting data form a hierarchical temporal sequence. Each three-year window contains monthly observations that are processed by the inner LSTM to learn seasonal ecological dynamics, whereas the sequence of ten overlapping windows is processed by the outer LSTM to capture long-term interannual trends. Spatial relationships among monitoring stations are simultaneously modeled by the GNN at each temporal step.

This representation provides several advantages over conventional year-by-year modeling. First, averaging over three years reduces sampling noise and zero inflation common in fisheries data while preserving long-term ecological signals. Second, the overlapping windows provide substantially richer temporal context than isolated annual observations, allowing the model to learn gradual ecological changes rather than abrupt year-to-year fluctuations. Third, because adjacent windows share two years of information, the representation generates a smooth temporal progression that improves learning efficiency and effectively augments the available training data without introducing synthetic observations. Finally, by separating seasonal variation

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(inner LSTM) from interannual dynamics (outer LSTM), the nested architecture simultaneously captures ecological processes operating across multiple temporal scales.

Importantly, the final window includes environmental and community predictors from the prediction year, representing information available at the time of prediction. The model is never provided with the observed abundance of the target year as input, thereby preventing information leakage while allowing predictions to be conditional on contemporaneous environmental conditions.

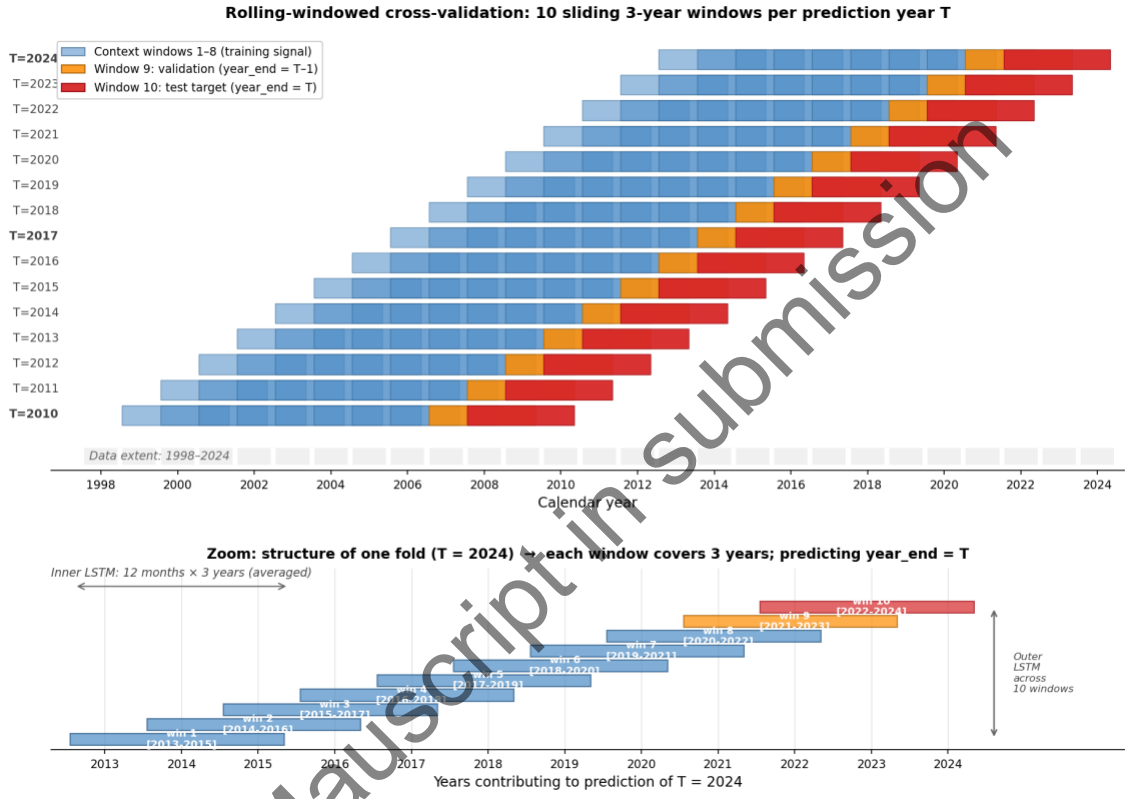



**Figure 3.** Rolling window approach for species data augmentation to predict time-series nekton population

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## 2.5. Model training objective

The training objective is represented by Equation 2. During model training, observed and predicted abundances were transformed back to a count scale using the inverse log transformation and then standardized on a per-species basis to account for differences in variance among species. Specifically, training was guided by an objective function that combines a robust abundance loss with Bayesian regularization:

$$L = L_{logcosh} + \beta_{KL} L_{KL} \quad (2)$$

$L_{logcosh}$ was used to quantify prediction error in the form of a numerically stable log-cosh loss (the logarithm of the hyperbolic cosine of the error; Equation 3):

$$L_{logcosh} = |x| + \log\left(1 + e^{-2|x|}\right) - \log(2) \quad (3)$$

where $x$ represents residual error. Log-cosh behaves similarly to mean squared error for small deviations and mean absolute error for large deviations, providing robustness to extreme values. This is particularly important for rare species, where standardized errors can be large and naive implementations of hyperbolic cosine loss may suffer from numerical instability. To further improve inference for rare species and regularize model complexity, a Bayesian Kullback–Leibler (KL) divergence weight $\beta_{KL} = 0.001$ was included (higher β = more regularization), penalizing deviations of species-specific posterior weight and bias distributions from their priors. The KL term was weighted by species-specific rarity factors ($L_{KL}$), increasing the contribution of rare species to the regularization and mitigating class imbalance and overfitting. During training, parameters were sampled from their variational posterior distributions using the reparameterization trick, enabling efficient gradient-based optimization, while posterior means were used at inference. Hereby, the model balances fitting the data accurately ($L_{logcosh}$) against keeping its internal representations well-behaved ($\beta_{KL} \cdot L_{KL}$).

The Bayesian variational heads handle both heterogeneous data density across species and a natural mechanism for knowledge transfer (ecologically similar species). That is, the KL divergence term automatically adjusts regularization strength per species and accepts informative priors when available. Although informative priors via cross-species learning are beneficial for data-limited species, they were unnecessary in this study because all target species had sufficient observations to learn species-specific parameters independently. Moreover, parameter sharing can obscure species-specific environmental responses by blending information across taxa, complicating the ecological interpretation of feature importance and partial dependence analyses. Because cross-species transfer provided no measurable improvement in predictive performance, we retained independent Bayesian output heads to maximize both predictive accuracy and ecological interpretability (Detailed analysis in Supplementary Materials).

We evaluated cross-donor Bayesian transfer learning, which uses the learned posterior of ecologically similar donor species as an informative prior for recipient species heads. As all seven target species (and their associated life history stages) are well-monitored with sufficient observations to independently learn species-specific parameters, the informative prior did not improve predictive accuracy over the uninformative baseline ($R^2$ = 0.602 vs. 0.605; detailed analysis in Supplementary Materials). The finalized model retains Bayesian variational heads with uninformative priors, which provide adaptive per-species regularization without cross-species transfer.

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## 2.6. Partial Dependence Analysis

Partial dependence plots (PDPs) (Friedman, 2001) were computed to visualize the marginal effect of individual environmental predictors on model-predicted abundance, averaging out the influence of all other features. The analysis used the same 2024 test-fold checkpoint and windowed data representation described previously (i.e., 10 windows, each a 3-year average).

We used two-dimensional partial dependence analyses to evaluate interactions between four key environmental predictors: water temperature (°C), salinity (PSU), dissolved oxygen (mg/L), and water depth (m; Friedman, 2001) and biotic community composition. Community groups were uniformly scaled from 0× to 2× baseline abundance to represent aggregate changes in prey or non-prey assemblages. We computed two-dimensional PDPs jointly for temperature,

salinity, dissolved oxygen, and prey/non-prey community. We selected Red Drum sub-adults and Spotted Seatrout recruits as targeted species-stages in 2-D PDP analysis because these represent the primary management-relevant life stages most closely associated with estuarine nursery habitats targeted by restoration.

For each target feature of water temperature, salinity, dissolved oxygen, and water depth, we constructed a grid of 20 evenly spaced values spanning the 5th–95th percentile of the normalized observed feature values across all stations and windows. At each grid point, the feature was fixed to that value across all stations and all windows while all other features retained their observed values.

Because prey and non-prey communities comprised many individual species, community effects were represented using a group-level scaling factor rather than separate PDPs for each taxon. For each community group, the abundances of all constituent species were simultaneously multiplied by a common factor ranging from 0 to 2, where 0× represented complete removal of the group, 1× represented observed baseline abundances, and 2× represented a twofold increase relative to baseline (e.g., 1.5× represented a hypothetical 50% increase in all constituent species). The focal environmental or habitat predictor was varied concurrently while other predictors were retained at their observed values, and predictions were averaged across observations. Full windowed inference was then run, and the mean predicted count was averaged across all stations and all 12 months of the held-out test year (2022) of the 2016-2020 rolling window to yield a single summary value per species per grid point and community assemblage level. These analyses therefore characterize model sensitivity to coordinated changes in aggregate community abundance and how those responses vary across environmental or habitat conditions.

## 2.7. Permutation-based feature group importance

The relative importance of ecological predictor groups was quantified using permutation-based feature importance, a model-agnostic approach that measures the dependence of model predictions on groups of correlated variables.

Model performance was first evaluated on the held-out test dataset to obtain the baseline mean squared error ($MSE_{base}$). One predictor group was then permuted at a time by randomly shuffling all variables belonging to that group across stations while leaving all remaining predictor groups unchanged. This procedure preserved the marginal distribution of each variable but disrupted the association between that predictor group and the response. The permuted dataset was passed through the trained model, and prediction error ($MSE_{perm,g}$) was recalculated.

Accuracy importance ($I_g$) for predictor group (g) was quantified as the relative increase in MSE after permutation (Equation 4):

$$I_g = \frac{\mathrm{MSE}_{\mathrm{perm,g}} - MSE_{base}}{MSE_{base}} \quad (4)$$

Positive $I_g$ values indicate that model performance deteriorated after permutation, indicating reliance on that predictor group for accurate prediction. Values near zero indicate little contribution to predictive performance, whereas negative values indicate that permutation slightly improved model performance, suggesting that the predictor group contributed little

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useful information or introduced noise. The permutation procedure was repeated five times for each predictor group, and importance values were averaged across repetitions.

In addition to prediction accuracy, the contribution of each predictor group to total predicted population abundance was evaluated. For each species, predicted abundance was summed across all stations and months to obtain a total predicted population under the baseline and permuted datasets. Absolute population importance (Equation 5) was calculated as the absolute difference between the baseline and permuted totals, averaged across species and permutation repeats:

$$\Delta P_g = \frac{1}{S}\sum_{i=1}^{S} \left| P_i^{base} - P_{i,g}^{perm} \right| \quad (5)$$

where $S$ is the number of species or species–life-stage groups, $P_s^{\text{base}}$ is the total predicted abundance of species $i$ under the original data, $P_{s,g}^{\text{perm}}$ is the total predicted abundance after permuting predictor group $g$. Larger values indicate that predictions are more sensitive to that predictor group. A signed version (i.e., positive or negative) of the population importance metric was also computed to determine the direction of each predictor group's effect on predicted abundance (Equation 6):

$$\Delta P_g^{signed} = \frac{1}{S}\sum_{i=1}^{S} \left( P_i^{base} - P_{i,g}^{perm} \right) \quad (6)$$

Positive values indicate that permuting the predictor group decreased predicted abundance, suggesting that the group contributed positively to population predictions. Negative values indicate that permuting the predictor group increased predicted abundance, suggesting that the group suppressed predicted abundance or was associated with lower predicted population estimates.

Permutation was performed at the predictor-group level rather than for individual variables because many ecological predictors are correlated within groups, such as multiple prey species or habitat descriptors. Group-wise permutation reduces compensation among correlated predictors and provides a more robust estimate of the contribution of each ecological predictor group to model performance and predicted abundance. Whereas values near zero indicate limited contribution to predictive performance.

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# 3. Results

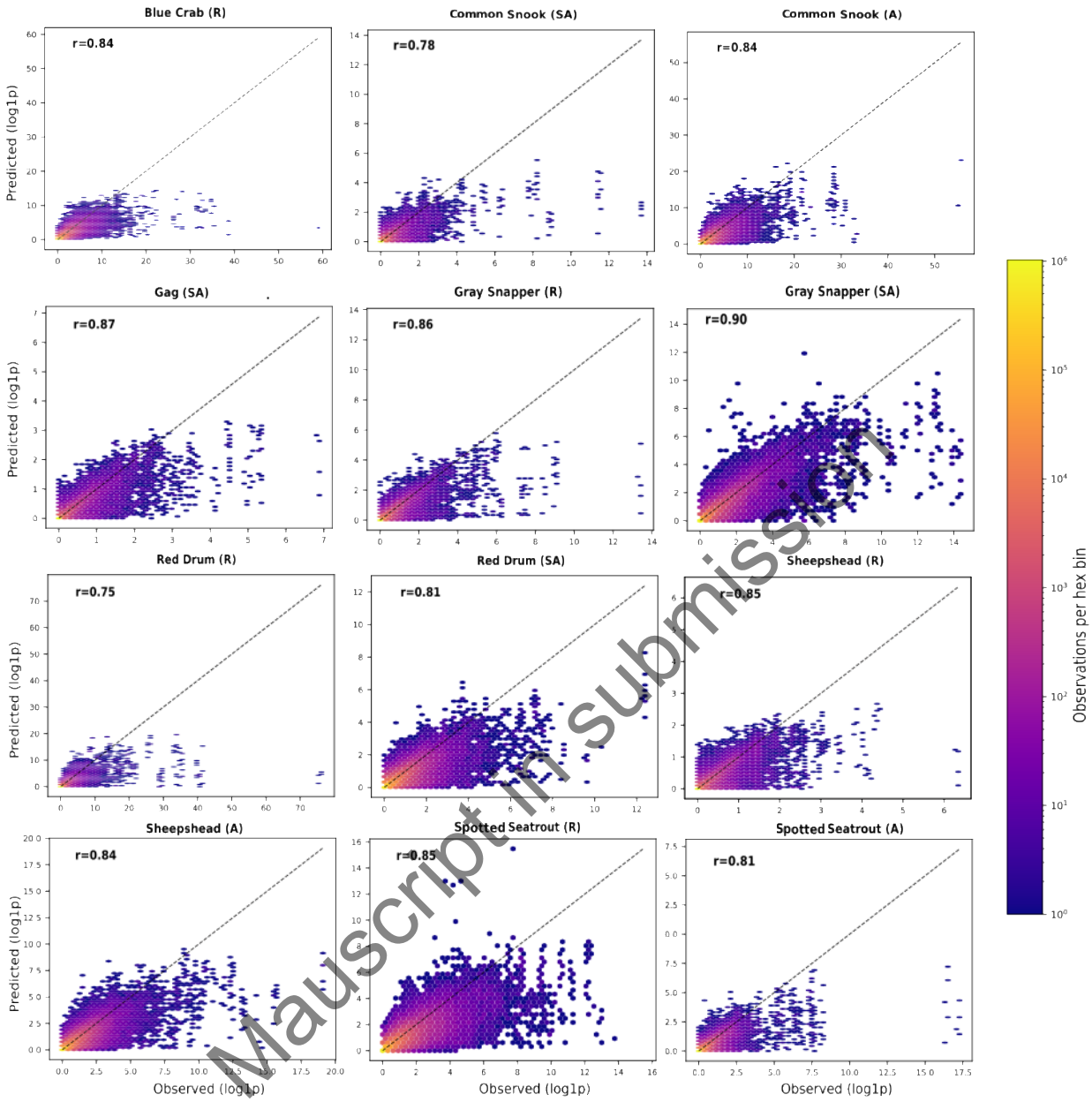
## 3.1. Model performance

Observed–predicted comparisons showed generally strong agreement across species and life stages, with observations concentrated around the 1:1 line (dotted black line) for most targets (Figure 4), indicating that the Bayesian GNN–nested LSTM model successfully captured long-term population dynamics over the 2010–2024 modeling period. Gray Snapper sub-adults, Gag Grouper sub-adults, and Gray Snapper recruits showed particularly compact distributions around the 1:1 relationship. In contrast, high-variance taxa such as Blue Crab recruits and adult Snook exhibited greater dispersion and systematic underprediction at high observed abundances.

Figure 4 further showed that prediction errors were concentrated primarily at abundance extremes, where the model tended to underestimate exceptionally high catches. For example, Red Drum recruits showed the greatest compression of predictions toward intermediate values, with reduced ability to reproduce extreme observations. Thus, the primary limitation was reproducing the magnitude of episodic abundance events rather than recovering broader patterns of population variation. Abundance was modeled on the log1p-transformed scale, as described in the Methods. Thus, substantial differences in raw catch at higher abundances appear comparatively smaller on the transformed scale, causing the model limited performance at high abundance points.

Figure 5 demonstrated that predictive performance varied among species and life stages, with $R^2$ ranging from approximately 0.55 to 0.80 and Pearson correlation coefficients ($r$) from 0.75 to 0.90. Despite this variation, the consistently positive and relatively high correlations indicated that the model captured broad spatial and temporal variation in abundance across all target species.

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**Figure 4.** Predicted and observed population by species on a log1p scale; the color map represents the number of observations. The dotted black line on each graph is the 1:1 line.

Gray Snapper sub-adults showed the highest performance ($R^2 = 0.80$, $r = 0.90$), and Red Drum recruits showed the lowest explained variance ($R^2 = 0.55$). RMSE provided complementary information because it was strongly influenced by the scale and variability of abundance: Blue Crab recruits and adult Common Snook had the largest RMSE despite relatively strong $R^2$ values (0.70), whereas lower-abundance taxa generally had smaller RMSE. Thus, high RMSE for abundant, variable taxa reflected larger absolute prediction errors rather than a failure to recover their overall spatiotemporal patterns (more specific metric values are provided in the Supplementary Materials Table S3).

A consistent pattern across species was the superior performance of subadult life stages relative to recruits. Subadult distributions are likely more strongly constrained by habitat characteristics and environmental gradients, resulting in greater predictability. In contrast, recruit abundances were more variable and generally exhibited lower R-squared values, particularly for Red Drum and Spotted Seatrout. This reduced predictability likely reflects recruitment pulses, larval transport, early-life mortality, and other stochastic demographic processes not fully represented by the environmental predictor set. Differences in predictive performance among life stages suggest a continuum between environmentally structured and demographically stochastic populations. Subadults with strong habitat associations were predicted with high accuracy. Recruit stages were less predictable, implying that environmental gradients explain a substantial portion of abundance variation for established individuals but a smaller fraction during early life stages dominated by recruitment dynamics.

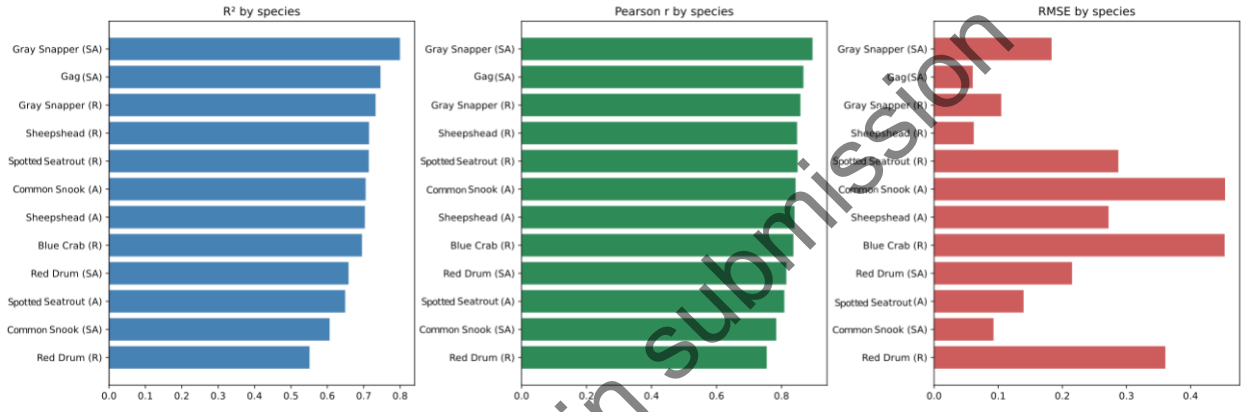


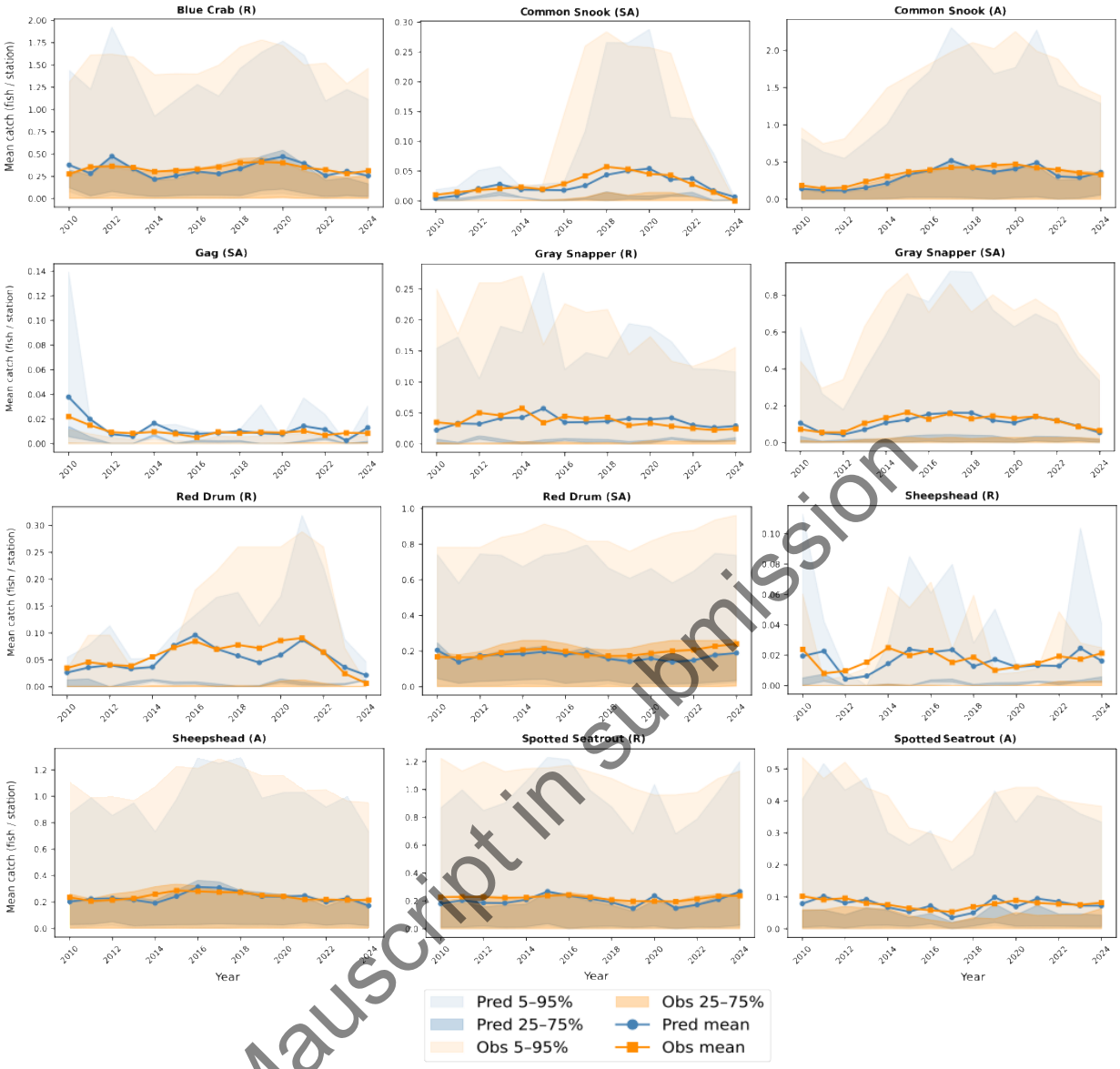

**Figure 5.** Prediction performance metrics by species.

## 3.2. Species population trends

Predicted trajectories closely tracked observed interannual fluctuations, including gradual increases and declines in abundance for most species (Figure 6). Deviations may be associated with years of exceptionally high recruitment, where the model tended to underestimate peak abundances. Most estuarine fish are r-selected, producing thousands to tens of thousands of offspring, the majority of which die before reaching maturity. Consequently, recruits are inherently highly variable and can appear in episodically high abundances driven by the timing of reproduction but may be encountered infrequently between these events. This stochastic variability in recruit abundance is difficult for any regression model to fully capture, as predictions are smoothed toward conditional means.

In contrast, subadult and adult populations, while also encountered infrequently, occur at consistently lower abundances and exhibit more stable interannual trajectories, resulting in correspondingly closer agreement between observed and predicted trends. Species with persistently low abundance, such as Gag subadults, displayed nearly flat temporal trajectories in both observations and predictions, confirming that the model accurately represented their rarity rather than overfitting to noise. The shaded bands represent the distribution of catches across sampling stations (25–75% and 5–95% quantiles), illustrating substantial spatial heterogeneity relative to changes in annual mean abundance. Despite this variability, the close agreement between observed and predicted mean trajectories demonstrates that the model successfully recovered long-term temporal population trends amid considerable spatial variation.

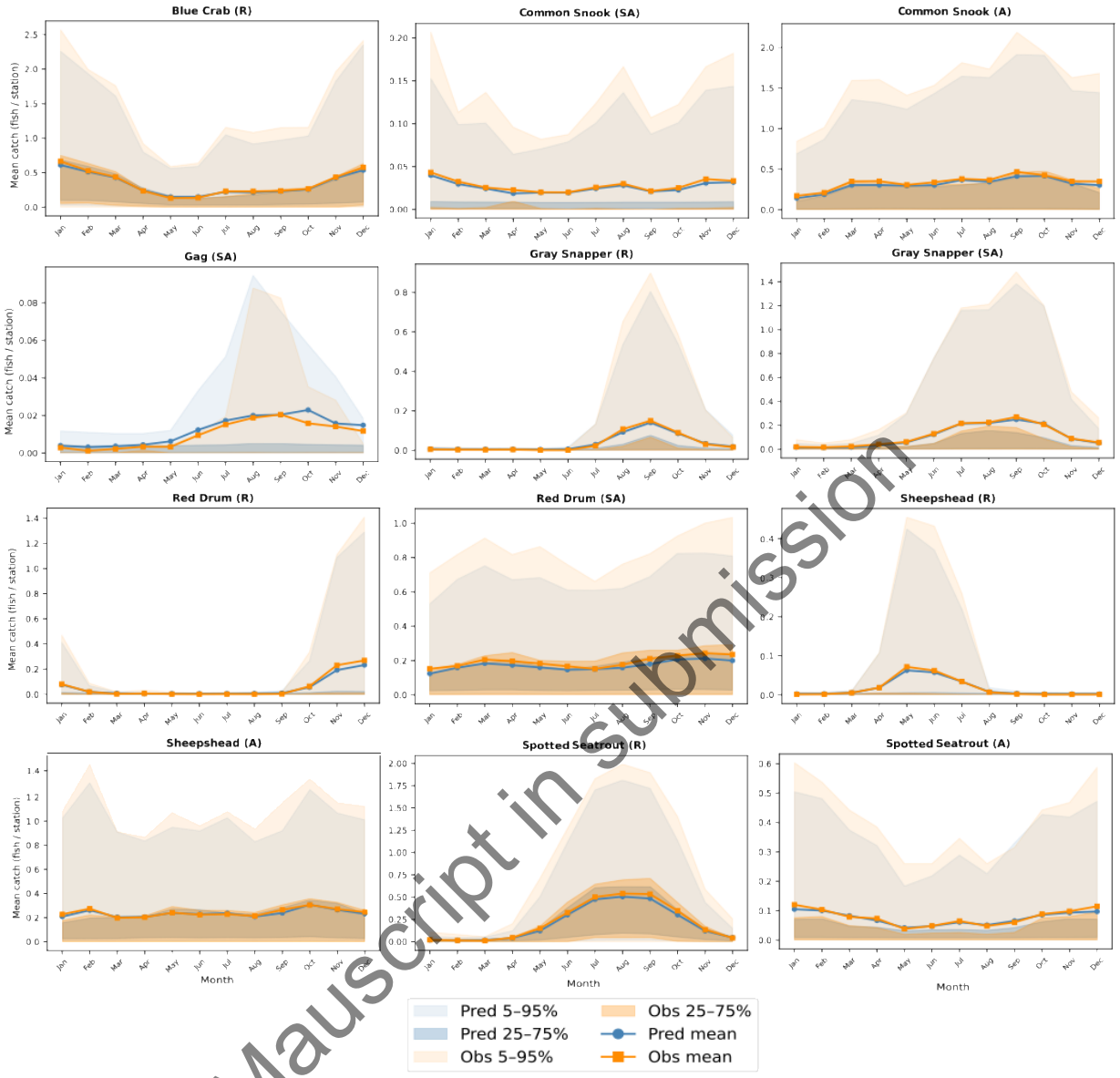
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**Figure 6.** Predicted and observed annual population trends (2010–2024). Solid lines represent the annual mean abundance across sampling stations for observed (blue) and predicted (orange) values. Dark and light shaded bands denote the interquartile (25–75%) and 5-95% ranged across sampling stations, respectively.

The model also reproduced species-specific seasonal phenology, with predicted monthly mean abundances closely matching observed seasonal cycles across most taxa (Figure 7). Recruit stages exhibited the strongest seasonal signatures, characterized by distinct periods of peak abundance, whereas adult stages generally displayed weaker seasonal fluctuations and greater temporal persistence throughout the year. These results demonstrate that the nested temporal architecture effectively captured both seasonal dynamics and interannual population trends operating at different temporal scales.




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**Figure 7.** Predicted and observed mean seasonal population cycles (monthly climatology averaged over 2010–2024). Solid lines represent the mean monthly abundance across sampling stations for observed (blue) and predicted (orange) values. Dark and light shaded ban

Prediction accuracy remained highest during periods of moderate abundance, whereas seasonal peaks associated with episodic recruitment were generally underestimated, particularly for species exhibiting recruitment pulses. This conservative behavior is consistent with regression models fitted to highly skewed abundance distributions, where extreme observations are relatively uncommon. Seasonal agreement was weaker for rare species such as Gag subadults, reflecting limited observations and greater stochasticity in seasonal occurrence. The wider shaded bands during peak months further indicate increased spatial heterogeneity among sampling stations when populations were most abundant.

Together, the annual and seasonal analyses demonstrate that the Bayesian GNN–nested LSTM model captures temporal dynamics across multiple ecological scales. The model

accurately reproduces long-term interannual population trajectories while simultaneously resolving recurring seasonal phenology across diverse species and life stages. Although exceptionally large recruitment events remain difficult to predict precisely, the model consistently captures the timing, direction, and relative magnitude of population change, indicating that the nested LSTM architecture effectively represents both long-term ecological trends and seasonal ecological processes.

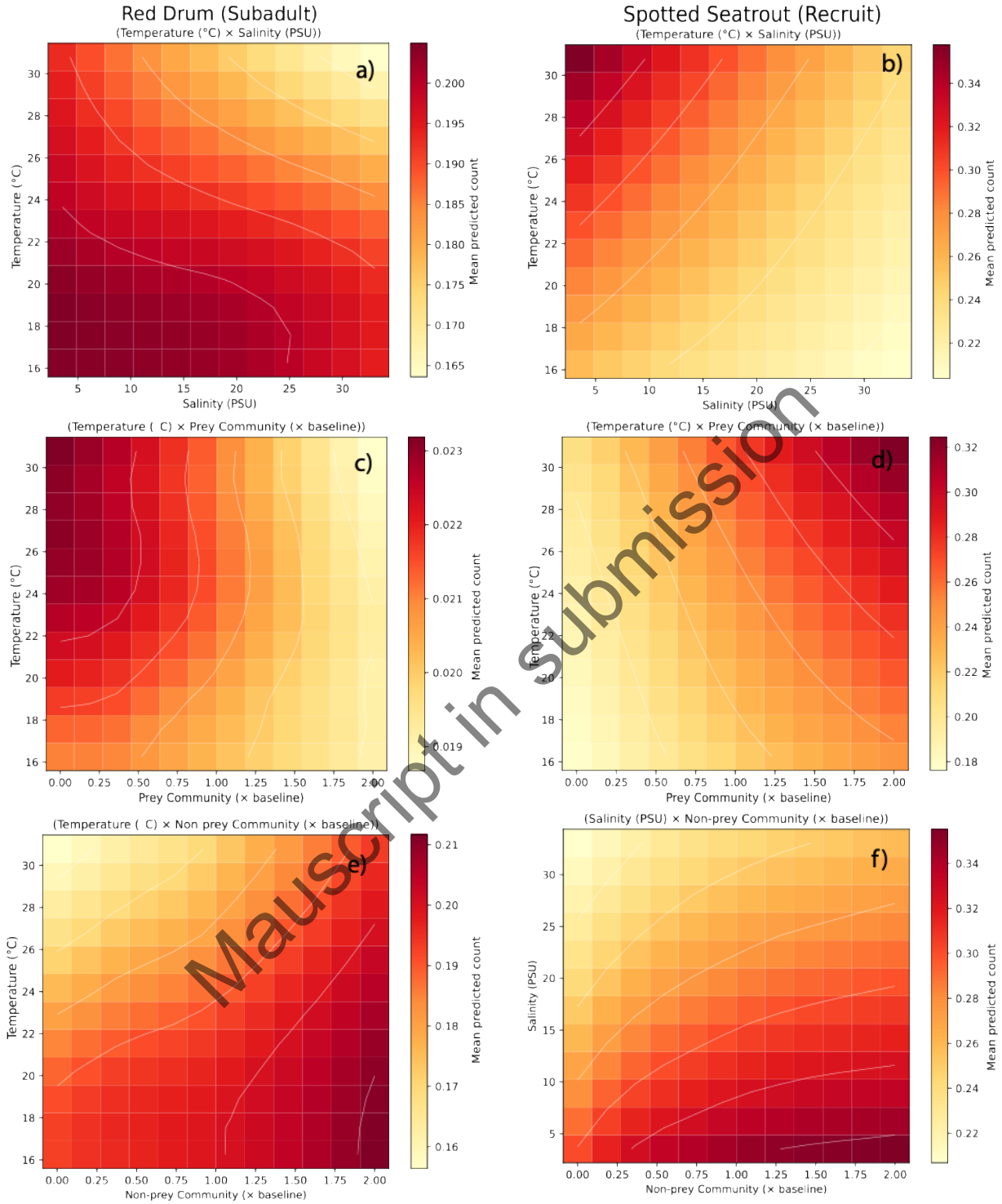
## 3.3. Joint abiotic and biotic responses

Two-dimensional partial dependence surfaces revealed distinct but context-dependent effects of abiotic conditions and community composition on predicted abundance (Figure 8). Both Red Drum sub-adults and Spotted Seatrout recruits showed higher predicted abundance under warm, low-salinity conditions, although the response was substantially stronger for Spotted Seatrout recruits. For Red Drum sub-adults, temperature and salinity exhibited a comparatively modest joint effect, whereas Spotted Seatrout recruits showed a pronounced gradient toward warm, low-salinity conditions, consistent with strong environmental structuring of recruitment habitat. Previous studies incorporating FIM data over similar geographical and salinity ranges also found that the occurrence of Spotted Seatrout recruits was negatively correlated with salinity at the time of sampling (Nelson and Leffler 2001; Flaherty-Walia et al. 2015). Likewise, Red Drum recruit to low-salinity habitats (Peters and McMichael 1987), but tend to become generalists as they become subadults.

Community effects also depend on environmental context. Predicted Red Drum sub-adult abundance increased with both prey and non-prey community abundance, but these associations varied across temperature, indicating that community composition and thermal conditions were not independent in the model response. For Spotted Seatrout recruits, prey-community abundance produced a strong positive response across much of the temperature range, whereas the association with non-prey community abundance was strongly dependent on salinity. In particular, increasing non-prey community abundance was associated with substantially greater predicted recruitment under low-salinity conditions, while this relationship weakened toward higher salinities.

Together, these surfaces indicate that biotic community composition modifies species responses to the abiotic environment rather than operating as an independent predictor of abundance. Red Drum sub-adults showed stronger context dependence between temperature and community composition, whereas Spotted Seatrout recruits exhibited particularly strong responses to community abundance within favorable low-salinity conditions. These results reinforce the broader attribution analysis by showing that the importance of prey and non-prey communities is environmentally contingent: favorable abiotic conditions and community context jointly characterize areas of high predicted abundance.

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**Figure 8.** Two-dimensional partial dependence surfaces for Red Drum sub-adults and Spotted Seatrout recruits (test year 2022). (a) Red Drum sub-adult: temperature × salinity. (b) Spotted Seatrout recruit: temperature × salinity. (c) Red Drum sub-adult: temperature × prey community abundance. (d) Spotted Seatrout recruit: temperature × prey community abundance. (e) Red Drum sub-adult: temperature × non-prey community abundance. (f) Spotted Seatrout recruit: salinity × non-prey community abundance. Color intensity represents mean predicted count (log1p scale); white contour lines indicate iso-response surfaces. Community abundance is expressed as a scaling factor relative to observed baseline (1×).

## 3.4. Relative importance of biotic and abiotic factors

Permutation analyses revealed complementary roles of predictor groups in determining prediction accuracy, overall population abundance (Figure 9), and species-specific responses (Figure 10). Community composition was the dominant predictor of model performance, with non-prey and prey community variables producing the largest increases in prediction error when permuted, substantially exceeding the contribution of all other predictor groups. This finding suggests that species assemblages contain substantial ecological information beyond that captured by the measured environmental and habitat variables. Survey conditions and covariance in catch by species provided moderate improvements in predictive accuracy, whereas submerged aquatic vegetation, abiotic environmental variables, and restoration contributed comparatively little to fine-scale spatial discrimination. In contrast, bottom cover, shoreline characteristics, and missing-data indicators had near-zero or slightly negative effects on prediction accuracy, indicating that these variables contributed little additional information for distinguishing where nekton occurred.

Predictor importance based on changes in total predicted abundance revealed a different hierarchy. Although prey and non-prey community variables remained the strongest predictors, shoreline characteristics, restoration, submerged aquatic vegetation, and abiotic environmental conditions produced substantially larger changes in total predicted abundance than would be expected from their relatively weak contributions to prediction accuracy. These results suggest that habitat-related predictors primarily regulate broad-scale habitat suitability and overall abundance levels, whereas community composition provides the fine-scale ecological context required to accurately resolve spatial and temporal variation in nekton distributions.

Species-specific signed permutation analyses further demonstrated that most predictor groups contributed positively to predicted abundance across taxa (Figure 11). The strongest positive responses occurred in the recruit stages of Blue Crab and Spotted Seatrout, and in subadult Red Drum, indicating that early life stages were the most sensitive to variation in community composition and habitat conditions. Shoreline characteristics and restoration exhibited their largest positive effects on juvenile and sub-adult stages, consistent with the importance of estuarine nursery habitats during early life-history stages. In contrast, negative responses were uncommon and largely restricted to individual adult species, suggesting that only a small number of habitat variables reduced predicted abundance for particular taxa.

In sum, these three analyses demonstrate that different predictor groups contribute to model performance through distinct ecological pathways. Community composition was the dominant determinant of prediction accuracy and remained the strongest predictor of overall abundance, whereas habitat-related variables primarily established broad-scale habitat suitability that modulated population size without substantially improving fine-scale spatial discrimination. The largest predictor effects occurred primarily during juvenile and subadult life stages, highlighting the greater environmental sensitivity of early developmental stages relative to adults.

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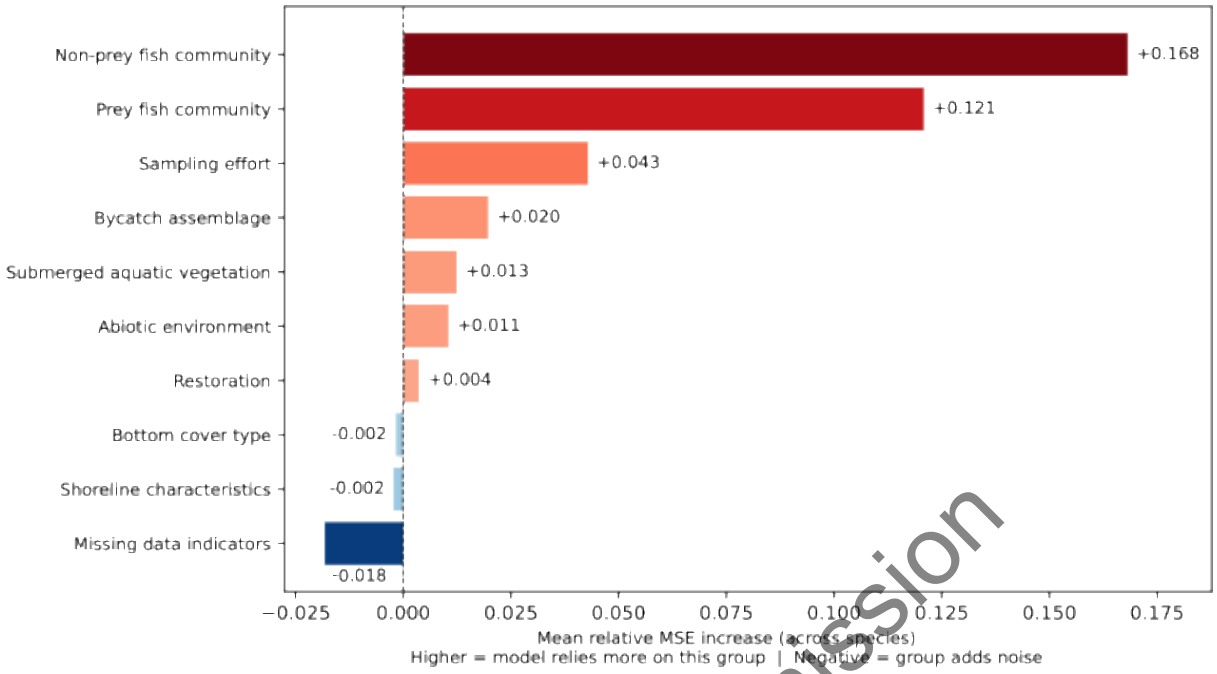



**Figure 9.** Feature group importance for prediction accuracy. Values represent the change in total mean squared error (MSE). Larger values indicate a greater influence of the predictor group on accuracy. A positive value indicates the model relies on this group for improving accuracy, and a negative value means this group adds noise to the model. Color intensity is proportional to the magnitude of the abundance change.

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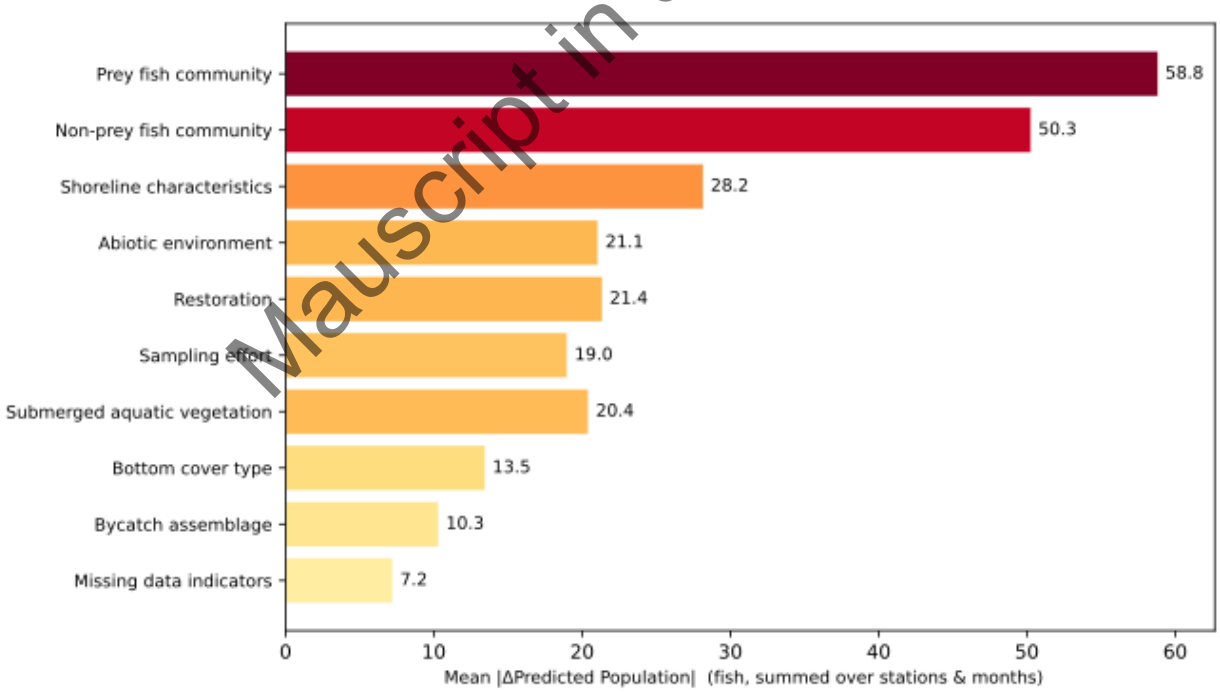



**Figure 10.** Feature group importance for prediction accuracy and predicted population abundance. Values represent the absolute change in total mean squared error (MSE) and predicted abundance after permuting each predictor group. Larger values indicate a greater influence of the predictor group on predicted population abundance. Color intensity is proportional to the magnitude of the abundance change.

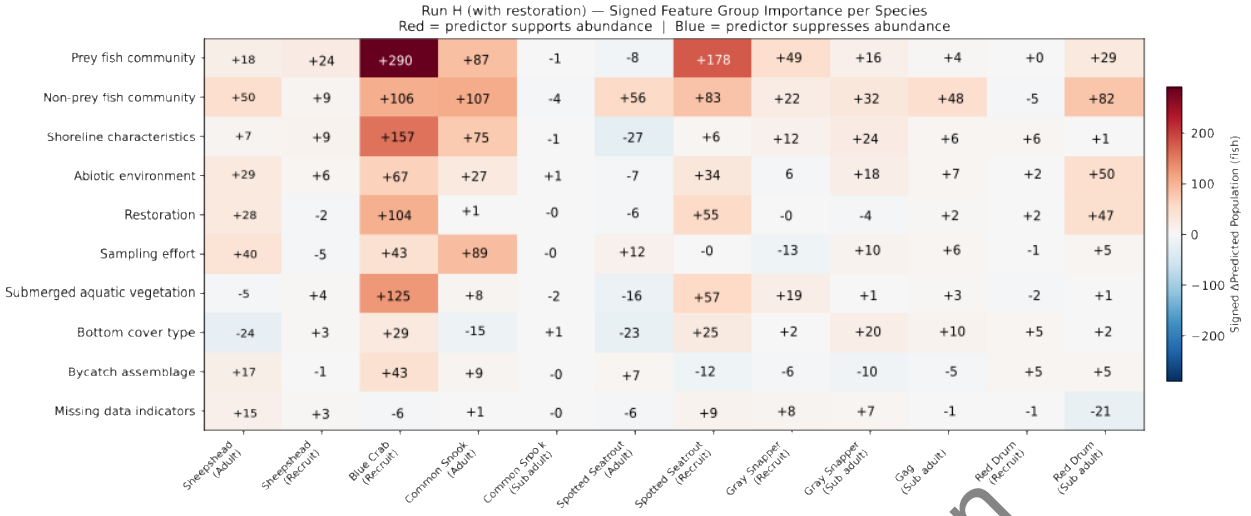



**Figure 11.** Importance of predictor groups for total predicted population abundance across species and life stages. Values represent the change in total predicted abundance after permuting each predictor group. Positive values indicate that including the predictor increased predicted abundance (positive contribution), whereas negative values indicate that including the predictor reduced predicted abundance (negative contribution). Color intensity is proportional to the magnitude of the abundance change. Y axis is order by descending predictor group importance

# 4. Discussion

Species distribution models have become important components of habit-based fisheries management. However, existing models lack the ability to synthesize and evaluate environmental conditions, biotic interactions, habitat characteristics, and management actions simultaneously. As a result, the relative importance of climatic and habitat variability, community composition, conservation and restoration efforts, and their interactive effects have remained poorly resolved. This limitation is particularly relevant in dynamic estuarine ecosystems, where these factors vary across space and through time. Recent advances, such as the Bayesian GNN–nLSTM described herein, help address these challenges by integrating diverse data within a unified framework, improving predictive capabilities and advancing our understanding of the factors that influence species abundance.

Differences in predictive performance among species and life stages likely reflect variation in the degree to which abundance is structured by deterministic environmental conditions versus stochastic demographic processes. In contrast, lower performance for recruit stages likely reflects the influence of episodic spawning, larval transport, early-life survival, and recruitment pulses that are not fully captured by the predictor set. This interpretation is consistent with broader ecological theory that species distributions emerge from the interaction of environmental filtering, biological interactions, and scale-dependent demographic processes (Levin, 1992; Planque et al., 2011).

Our permutation-based analyses show that biotic community composition was the strongest contributor to predictive accuracy. Traditional SDMs have often emphasized abiotic environmental gradients, particularly temperature, salinity, depth, and other habitat variables, as primary predictors of species distributions. However, increasing evidence shows that biotic interactions can substantially shape realized distributions and community composition, and that ignoring these interactions may limit ecological inference (Wisz et al., 2013; Dormann et al.,

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2018; Poggiato et al., 2021). Our results support that prey and non-prey community predictors contribute more strongly to accurate spatial and temporal prediction than abiotic or habitat variables alone. Our $R^2$ values represent a substantial improvement over previous regression-based methods for predicting FIM inshore fish abundance. Ainsworth et al. (2025) used generalized additive models incorporating *in situ* and regional environmental predictors to model a similar set of species. As in the present study, Gag and other Serranidae were among the most predictable taxa based on environmental conditions; however, predictive performance was considerably lower in their study, with pseudo-$R^2$ values of 0.21 for Gag and 0.36 for other Serranidae. Smaller-bodied species generally exhibited much lower predictability, and much of the remaining unexplained variance remained in a Year term, suggesting the presence of important environmental or ecological drivers that were not represented in the predictor set. This parallels the results in the current study which suggest that a portion of this remaining environmental signal is captured in the biotic community information, which emerged as a strong contributor to predictive performance.

At the same time, habitat and management variables played a distinct role. Although shoreline characteristics, submerged aquatic vegetation, and restoration predictors contributed relatively little to fine-scale prediction accuracy, they had substantially larger effects on total predicted abundance. This contrast suggests that biotic community composition helps identify where nekton occur, whereas habitat and management predictors influence how many individuals the system can support. These finding complements habitat-based fisheries modeling studies in the Gulf of Mexico, which have shown that environmental and benthic habitat variables improve spatial distribution estimates for fish and invertebrates (Drexler & Ainsworth, 2013; Grüss et al., 2014, 2018, 2019; Rubec et al., 2016). Our work builds on these studies by integrating biotic, abiotic, and restoration predictors within a single model and by explicitly separating predictors that improve spatial accuracy from those that affect population magnitude.

The relatively modest contribution of restoration variables to prediction accuracy should not be interpreted as evidence that restoration is ineffective. The fact that restoration and habitat variables affected predicted population abundance more than predictive accuracy suggests that restoration may operate primarily by modifying broad-scale habitat suitability or abundance potential rather than by improving fine-scale spatial discrimination among sampling locations.

Because the Bayesian GNN–nLSTM simultaneously accounts for environmental variability, species interactions, and management actions, it provides a powerful framework for separating restoration effects from confounding climatic and ecological drivers. Future applications of this framework could identify which restoration strategies yield the greatest ecological benefit and support adaptive restoration planning amid ongoing climate change.

A practical limitation of deep learning approaches in applied ecology is the technical barrier to adoption and reproducibility. Unlike generalized linear or additive models, which are accessible through widely used statistical software (e.g., R packages mgcv (Wood, 2017), glmmTMB (Brooks et al., 2017), the graph neural network in our model requires familiarity with specialized frameworks such as PyTorch (Paszke et al., 2019). However, the present model's computational demands are modest because training and inference run on standard hardware without dedicated GPU resources, reducing a common barrier to reproducibility. To further support accessibility, we provide all model code, trained weights, and preprocessing pipelines under an open-source license. Broader adoption will also benefit from continued development of accessible deep learning interfaces for ecological applications (e.g., Christin et al., 2019;

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Borowiec et al., 2022) and from cross-disciplinary collaboration between ecologists and data scientists.

# 5. Conclusion

Using a Bayesian GNN–nLSTM framework, we found that biotic community composition was the strongest determinant of predictive accuracy, indicating that species (nekton) distributions are governed more by species interactions than by environmental conditions alone. Notably, the prey and non-prey community structure was the dominant contributor to predictive accuracy, indicating that nekton population recovery depends strongly on species co-occurrence and ecological interactions. In contrast, habitat- and management-related variables, including shoreline characteristics, restoration activities, submerged aquatic vegetation, and other habitat attributes, had stronger effects on predicted population abundance than on spatial prediction accuracy. These findings suggest that the processes governing where nekton occur differ from those governing how many individuals can be supported, highlighting complementary roles for community interactions and habitat management in ecosystem recovery.

Methodologically, our framework addresses key challenges in ecological modeling by simultaneously accounting for spatial and temporal autocorrelation, improving predictions for rare species through Bayesian transfer learning, integrating environmental, habitat, biotic, and management predictors within a unified framework, and disentangling their relative contributions through interpretable permutation attribution. Although developed using long-term fisheries-independent monitoring data from Florida estuaries, the Bayesian GNN–nLSTM is broadly transferable to other species, ecosystems, and geographic regions, providing a general framework for ecological forecasting, restoration prioritization, and ecosystem-based conservation under global environmental change.

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# 6. Acknowledgement

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# 7. Code and data accessibility

Code and data are available on Zenodo at 10.5281/zenodo.22213363

# 8. References

Ahmadian, Mahshid, Edward L. Boone, and Grace S. Chiu. “A Methodology for Spatiotemporal Modeling of Acoustically Tagged Fish Data.” *arXiv Preprint arXiv:2408.13220*, 2024.

Ahmadian, Mahshid, Edward L. Boone, Grace S. Chiu, and Kevin C. Weng. "A New Perspective to Fish Trajectory Imputation: A Spatiotemporal Probability Model for Simulating Acoustically Tagged Fish Movement." *Journal of Applied Statistics: Environmental Statistics and Data Science* 2, no. 1 (2026): 2623273. https://doi.org/10.1080/29984688.2026.2623273.

Ainsworth, Cameron H., Hallie C. Repeta, and Rebecca L. Scott. "You Can't Reach the Lost Valley by Boat: Navigating Bottom-Up Restoration Pathways." *Fish and Fisheries* 26, no. 4 (2025): 546–60. https://doi.org/10.1111/faf.12897.

Bayraktarov, Elisa, Megan I. Saunders, Sabah Abdullah, et al. "The Cost and Feasibility of Marine Coastal Restoration." *Ecological Applications* 26, no. 4 (2016): 1055–74. https://doi.org/10.1890/15-1077.

Beck, Michael W., Kenneth L. Heck, Kenneth W. Able, et al. "The Identification, Conservation, and Management of Estuarine and Marine Nurseries for Fish and Invertebrates." *BioScience* 51, no. 8 (2001): 633. https://doi.org/10.1641/0006-3568(2001)051%255B0633:TICAMO%255D2.0.CO;2.

Borowiec, Marek L., Rebecca B. Dikow, Paul B. Frandsen, Alexander McKeeken, Gabriele Valentini, and Alexander E. White. "Deep Learning as a Tool for Ecology and Evolution." *Methods in Ecology and Evolution* 13, no. 8 (2022): 1640–60. https://doi.org/10.1111/2041-210X.13901.

Bosso, Luciano, Raffaele Panzuto, Rosario Balestrieri, et al. "Integrating Citizen Science and Spatial Ecology to Inform Management and Conservation of the Italian Seahorses." *Ecological Informatics* 79 (March 2024): 102402. https://doi.org/10.1016/j.ecoinf.2023.102402.

Boström, C., Sj Pittman, C. Simenstad, and Rt Kneib. "Seascape Ecology of Coastal Biogenic Habitats: Advances, Gaps, and Challenges." *Marine Ecology Progress Series* 427 (April 2011): 191–217. https://doi.org/10.3354/meps09051.

Briggs, John C. "A List of Florida Fishes and Their Distribution." *Bulletin of the Florida Museum of Natural History* 2, no. 8 (1958): 223–318. https://doi.org/10.58782/flmnh.iwxo3542.

Brooks, Mollie, E., Kasper Kristensen, Koen Benthem J. ,van, et al. "glmmTMB Balances Speed and Flexibility Among Packages for Zero-Inflated Generalized Linear Mixed Modeling." *The R Journal* 9, no. 2 (2017): 378. https://doi.org/10.32614/RJ-2017-066.

Caron, Mathilde, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, and Armand Joulin. "Unsupervised Learning of Visual Features by Contrasting Cluster Assignments." arXiv:2006.09882. Preprint, arXiv, January 8, 2021. https://doi.org/10.48550/arXiv.2006.09882.

Cherubini, Carla, Leonardo Saccotelli, Giulia Cipriano, et al. "Enhancing Marine Ecosystem Monitoring Using Machine Learning: Addressing Autocorrelation in Cetacean Feeding Behavior Prediction." In *Multimodal Sensing and Artificial Intelligence for Sustainable Future*, edited by Claas Falldorf, Francesco Soldovieri, Vittorio Bianco, and Pascal Picart. SPIE, 2025. https://doi.org/10.1117/12.3062413.

Cheung, William W. L., Vicky W. Y. Lam, Jorge L. Sarmiento, et al. "Large-Scale Redistribution of Maximum Fisheries Catch Potential in the Global Ocean under Climate Change." *Global Change Biology* 16, no. 1 (2010): 24–35. https://doi.org/10.1111/j.1365-2486.2009.01995.x.

Manuscript in submission

680 Christin, Sylvain, Éric Hervet, and Nicolas Lecomte. “Applications for Deep Learning in
681 Ecology.” *Methods in Ecology and Evolution* 10, no. 10 (2019): 1632–44.
682 https://doi.org/10.1111/2041-210X.13256.
683 Darnell, Kelly M., M. Zachary Darnell, D. Lee Smee, Charles W. Martin, Margaret O. Hall, and
684 Bradley Furman. “NOAA RESTORE Science Program: Gulf-Wide Assessment of Habitat
685 Use and Habitat-Specific Production Estimates of Nekton in Turtlegrass (Thalassia
686 Testudinum): Nekton and Primary Producer Stable Isotopes in Turtlegrass-Dominated
687 Seagrass Beds in the Northern Gulf of America, 2018-05-23 to 2018-10-03 (NCEI
688 Accession 0304846).” NOAA National Centers for Environmental Information, 2025.
689 https://doi.org/10.25921/9B8J-ZN57.
690 Diaz, Robert J., and Rutger Rosenberg. “Spreading Dead Zones and Consequences for Marine
691 Ecosystems.” *Science* 321, no. 5891 (2008): 926–29.
692 https://doi.org/10.1126/science.1156401.
693 Dormann, Carsten F., Maria Bobrowski, D. Matthias Dehling, et al. “Biotic Interactions in
694 Species Distribution Modelling: 10 Questions to Guide Interpretation and Avoid False
695 Conclusions.” *Global Ecology and Biogeography* 27, no. 9 (2018): 1004–16.
696 https://doi.org/10.1111/geb.12759.
697 Drexler, Michael D., and Cameron H. Ainsworth. “Generalized Additive Models Used to Predict
698 Species Abundance in the Gulf of Mexico: An Ecosystem Modeling Tool.” *PLoS ONE* 8,
699 no. 5 (2013): e64458. https://doi.org/10.1371/journal.pone.0064458.
700 Fan, Joshua, Junwen Bai, Zhiyun Li, Ariel Ortiz-Bobea, and Carla P. Gomes. “A GNN-RNN
701 Approach for Harnessing Geospatial and Temporal Information: Application to Crop Yield
702 Prediction.” *NeurIPS Workshop on Tackling Climate Change with Machine Learning*, 2021.
703 Fan, Joshua, Junwen Bai, Zhiyun Li, Ariel Ortiz-Bobea, and Carla P. Gomes. “A GNN-RNN
704 Approach for Harnessing Geospatial and Temporal Information: Application to Crop Yield
705 Prediction.” Preprint, arXiv, 2021. https://doi.org/10.48550/ARXIV.2111.08900.
706 Flaherty, Kerry E., Theodore S. Switzer, Brent L. Winner, and Sean F. Keenan. “Regional
707 Correspondence in Habitat Occupancy by Gray Snapper (*Lutjanus Griseus*) in Estuaries of
708 the Southeastern United States.” *Estuaries and Coasts* 37, no. 1 (2014): 206–28.
709 https://doi.org/10.1007/s12237-013-9652-x.
710 Flaherty-Walia, Kerry E., Richard E. Matheson, and Richard Paperno. “Juvenile Spotted
711 Seatrout (Cynoscion Nebulosus) Habitat Use in an Eastern Gulf of Mexico Estuary: The
712 Effects of Seagrass Bed Architecture, Seagrass Species Composition, and Varying Degrees
713 of Freshwater Influence.” *Estuaries and Coasts* 38, no. 1 (2015): 353–66.
714 https://doi.org/10.1007/s12237-014-9821-6.
715 Friedman, Jerome H. “Greedy Function Approximation: A Gradient Boosting Machine.” *The*
716 *Annals of Statistics* 29, no. 5 (2001). https://doi.org/10.1214/aos/1013203451.
717 FWC-FWRI. “Fisheries-Independent Monitoring Program Procedure Manual.” St. Petersburg,
718 Florida, 2023.
719 Gillanders, Bronwyn M., Travis S. Elsdon, Ian A. Halliday, Gregory P. Jenkins, Julie B. Robins,
720 and Fiona J. Valesini. “Potential Effects of Climate Change on Australian Estuaries and
721 Fish Utilising Estuaries: A Review.” *Marine and Freshwater Research* 62, no. 9 (2011):
722 1115. https://doi.org/10.1071/MF11047.
723 Grüss, Arnaud, Michael D. Drexler, and Cameron H. Ainsworth. “Using Delta Generalized
724 Additive Models to Produce Distribution Maps for Spatially Explicit Ecosystem Models.”
725 *Fisheries Research* 159 (2014): 11–24. https://doi.org/10.1016/j.fishres.2014.05.005.

Manuscript in Submission

Grüss, Arnaud, Michael D. Drexler, Cameron H. Ainsworth, Elizabeth A. Babcock, Joseph H. Tarnecki, and Matthew S. Love. "Producing Distribution Maps for a Spatially-Explicit Ecosystem Model Using Large Monitoring and Environmental Databases and a Combination of Interpolation and Extrapolation." *Frontiers in Marine Science* 5 (January 2018): 16. https://doi.org/10.3389/fmars.2018.00016.

Grüss, Arnaud, Michael D. Drexler, Emily Chancellor, et al. "Representing Species Distributions in Spatially-Explicit Ecosystem Models from Presence-Only Data." *Fisheries Research* 210 (February 2019): 89–105. https://doi.org/10.1016/j.fishres.2018.10.011.

Hale, Robin, Ralph Mac Nally, Daniel T. Blumstein, and Stephen E. Swearer. "Evaluating Where and How Habitat Restoration Is Undertaken for Animals." *Restoration Ecology* 27, no. 4 (2019): 775–81. https://doi.org/10.1111/rec.12958.

Halpern, Benjamin S., Catherine Longo, Darren Hardy, et al. "An Index to Assess the Health and Benefits of the Global Ocean." *Nature* 488, no. 7413 (2012): 615–20. https://doi.org/10.1038/nature11397.

Harrell, Lauren, Christine Kaeser-Chen, Burcu Karagol Ayan, et al. *Heterogeneous Graph Neural Networks for Species Distribution Modeling*. 2025. https://doi.org/10.48550/ARXIV.2503.11900.

Hayes, Ct, Sb Alford, Ba Belgrad, et al. "Regional Variation in Seagrass Complexity Drives Blue Crab *Callinectes Sapidus* Mortality and Growth across the Northern Gulf of Mexico." *Marine Ecology Progress Series* 693 (July 2022): 141–55. https://doi.org/10.3354/meps14084.

He, Kaiming, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. "Momentum Contrast for Unsupervised Visual Representation Learning." *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020, 9726–35. https://doi.org/10.1109/CVPR42600.2020.00975.

Heck, Kenneth L., Jr., G. Hays, and Robert J. Orth. "Critical Evaluation of the Nursery Role Hypothesis for Seagrass Meadows." *Marine Ecology Progress Series* 253 (2003): 123–36. https://doi.org/10.3354/meps253123.

Hovel, Ka, Ms Fonseca, Dl Myer, Wj Kenworthy, and Pe Whitfield. "Effects of Seagrass Landscape Structure, Structural Complexity and Hydrodynamic Regime on Macrofaunal Densities in North Carolina Seagrass Beds." *Marine Ecology Progress Series* 243 (2002): 11–24. https://doi.org/10.3354/meps243011.

Jeung, Minhyuk, Min-Chul Jang, Kyoungsoon Shin, Seung Won Jung, and Sang-Soo Baek. "Graph Neural Networks and Transfer Entropy Enhance Forecasting of Mesozooplankton Community Dynamics." *Environmental Science and Ecotechnology* 23 (January 2025): 100514. https://doi.org/10.1016/j.ese.2024.100514.

Kirwan, Matthew L., and J. Patrick Megonigal. "Tidal Wetland Stability in the Face of Human Impacts and Sea-Level Rise." *Nature* 504, no. 7478 (2013): 53–60. https://doi.org/10.1038/nature12856.

Lester, Sarah E., Benjamin S. Halpern, Kirsten Grorud-Colvert, et al. "Biological Effects within No-Take Marine Reserves: A Global Synthesis." *Marine Ecology Progress Series* 384 (May 2009): 33–46. https://doi.org/10.3354/meps08029.

Levin, Simon A. "The Problem of Pattern and Scale in Ecology: The Robert H. MacArthur Award Lecture." *Ecology* 73, no. 6 (1992): 1943–67. https://doi.org/10.2307/1941447.

Li, Liying, Junwen Bai, Shoukun Sun, Marcos Abel Zuzuárregui, and Zhe Wang. "The Risk of Birds' Functional Diversity Loss with Climate Change Uncovered Using Deep Learning

Manuscript in Submission

Hurdle Models." *Ecological Indicators* 190 (September 2026): 115321. https://doi.org/10.1016/j.ecolind.2026.115321.
Livingston, Robert J. "Resource Utilization as a Function of Restoring Freshwater Flow to the Apalachicola River Estuary." *Estuaries* 6 (1983): 311–21.
Lotze, Heike K., Hunter S. Lenihan, Bruce J. Bourque, et al. "Depletion, Degradation, and Recovery Potential of Estuaries and Coastal Seas." *Science* 312, no. 5781 (2006): 1806–9. https://doi.org/10.1126/science.1128035.
McCune, Bruce, James B. Grace, and Urban, Dean L. *Analysis of Ecological Communities*. 2nd printing. MjM Software Design, 2002.
McEachran, John D., and Janice D. Fechhelm. *Fishes of the Gulf of Mexico, Vol. 1: Myxiniformes to Gasterosteiformes*. Vol. 1. University of Texas Press, 1998.
McMichael, R. H. "Florida's Marine Fisheries-Independent Monitoring Program." In *Proceedings, Tampa Bay Area Scientific Information Symposium 2 (BASIS 2)*, edited by S. F. Treat and P. A. Clark. Tampa Bay Regional Planning Council, 1991.
Nagelkerken, Ivan, Marcus Sheaves, Ronald Baker, and Rod M. Connolly. "The Seascape Nursery: A Novel Spatial Approach to Identify and Manage Nurseries for Coastal Marine Fauna." *Fish and Fisheries* 16, no. 2 (2015): 362–71. https://doi.org/10.1111/faf.12057.
Nelson, Gary A., and Deborah Leffler. "Abundance, Spatial Distribution, and Mortality of Young-of-the-Year Spotted Seatrout (Cynoscion Nebulosus) Along the Gulf Coast of Florida." *Gulf of Mexico Science* 19, no. 1 (2001). https://doi.org/10.18785/goms.1901.03.
Nicholson, William R. "Movements and Population Structure of Atlantic Menhaden Indicated by Tag Returns." *Estuaries* 1, no. 3 (1978): 141. https://doi.org/10.2307/1351455.
Orth, Robert J., Tim J. B. Carruthers, William C. Dennison, et al. "A Global Crisis for Seagrass Ecosystems." *BioScience* 56, no. 12 (2006): 987. https://doi.org/10.1641/0006-3568(2006)56%255B987:AGCFSE%255D2.0.CO;2.
Osland, Michael J., Richard H. Day, Courtney T. Hall, Marisa D. Brumfield, Jason L. Dugas, and William R. Jones. "Mangrove Expansion and Contraction at a Poleward Range Limit: Climate Extremes and Land-ocean Temperature Gradients." *Ecology* 98, no. 1 (2017): 125–37. https://doi.org/10.1002/ecy.1625.
Ovaskainen, Otso, Steven Winter, Gleb Tikhonov, et al. "Common to Rare Transfer Learning (CORAL) Enables Inference and Prediction for a Quarter Million Rare Malagasy Arthropods." *Nature Methods* 22, no. 10 (2025): 2074–82. https://doi.org/10.1038/s41592-025-02823-y.
Peters, Kevin M., and Robert H. McMichael. "Early Life History of the Red Drum, Sciaenops Ocellatus (Pisces: Sciaenidae), in Tampa Bay, Florida." *Estuaries* 10, no. 2 (1987): 92. https://doi.org/10.2307/1352173.
Pinsky, Malin L., Boris Worm, Michael J. Fogarty, Jorge L. Sarmiento, and Simon A. Levin. "Marine Taxa Track Local Climate Velocities." *Science* 341, no. 6151 (2013): 1239–42. https://doi.org/10.1126/science.1239352.
Planque, Benjamin, Christophe Loots, Pierre Petitgas, Ulf LINDSTRøM, and Sandrine Vaz. "Understanding What Controls the Spatial Distribution of Fish Populations Using a Multi-model Approach." *Fisheries Oceanography* 20, no. 1 (2011): 1–17. https://doi.org/10.1111/j.1365-2419.2010.00546.x.
Poggiato, Giovanni, Tamara Münkemüller, Daria Bystrova, Julyan Arbel, James S. Clark, and Wilfried Thuiller. "On the Interpretations of Joint Modeling in Community Ecology."

Manuscript in Submission

*Trends in Ecology & Evolution* 36, no. 5 (2021): 391–401. https://doi.org/10.1016/j.tree.2021.01.002.

Poloczanska, Elvira S., Christopher J. Brown, William J. Sydeman, et al. "Global Imprint of Climate Change on Marine Life." *Nature Climate Change* 3, no. 10 (2013): 919–25. https://doi.org/10.1038/nclimate1958.

Pörtner, Hans O., and Anthony P. Farrell. "Physiology and Climate Change." *Science* 322, no. 5902 (2008): 690–92. https://doi.org/10.1126/science.1163156.

Rees, Matthew J., Nathan A. Knott, and Andrew R. Davis. "Habitat and Seascape Patterns Drive Spatial Variability in Temperate Fish Assemblages: Implications for Marine Protected Areas." *Marine Ecology Progress Series* 607 (December 2018): 171–86. https://doi.org/10.3354/meps12790.

Reichstein, Markus, Gustau Camps-Valls, Bjorn Stevens, et al. "Deep Learning and Process Understanding for Data-Driven Earth System Science." *Nature* 566, no. 7743 (2019): 195–204. https://doi.org/10.1038/s41586-019-0912-1.

Rubec, Peter J., Jesse Lewis, David Reed, et al. "Linking Oceanographic Modeling and Benthic Mapping with Habitat Suitability Models for Pink Shrimp on the West Florida Shelf." *Marine and Coastal Fisheries* 8, no. 1 (2016): 160–76. https://doi.org/10.1080/19425120.2015.1082519.

Rydene, David A., and Richard E. Matheson. "Diurnal Fish Density in Relation to Seagrass and Drift Algae Cover in Tampa Bay, Florida." *Gulf of Mexico Science* 21, no. 1 (2003). https://doi.org/10.18785/goms.2101.04.

Schmidt, Nancy, and Mark E. Luther. "ENSO Impacts on Salinity in Tampa Bay, Florida." *Estuaries* 25, no. 5 (2002): 976–84. https://doi.org/10.1007/BF02691345.

Schrandt, Meagan N., and Timothy C. MacDonald. "Long-Term Stability of the Faunal Community of a Subtropical Estuary: Evaluating Disturbances in the Context of Interannual Variability." *Estuaries and Coasts* 43, no. 2 (2020): 347–59. https://doi.org/10.1007/s12237-019-00684-1.

Stoner, Allan W., and Mary L. Ottmar. "Fish-Habitat Relationships and the Role of Seagrass Landscapes." *Estuaries* 27 (2004): 836–47.

Switzer, Theodore S., Timothy C. MacDonald, Robert H. McMichael, and Sean F. Keenan. "Recruitment of Juvenile Gags in the Eastern Gulf of Mexico and Factors Contributing to Observed Spatial and Temporal Patterns of Estuarine Occupancy." *Transactions of the American Fisheries Society* 141, no. 3 (2012): 707–19. https://doi.org/10.1080/00028487.2012.675913.

Toma, Tiago Shizen Pacheco, Gerhard Ernst Overbeck, Milton De Souza Mendonça, and G. Wilson Fernandes. "Optimal References for Ecological Restoration: The Need to Protect References in the Tropics." *Perspectives in Ecology and Conservation* 21, no. 1 (2023): 25–32. https://doi.org/10.1016/j.pecon.2023.01.003.

Waycott, Michelle, Carlos M. Duarte, Tim J. B. Carruthers, et al. "Accelerating Loss of Seagrasses across the Globe Threatens Coastal Ecosystems." *Proceedings of the National Academy of Sciences* 106, no. 30 (2009): 12377–81. https://doi.org/10.1073/pnas.0905620106.

Wisz, Mary Susanne, Julien Pottier, W. Daniel Kissling, et al. "The Role of Biotic Interactions in Shaping Distributions and Realised Assemblages of Species: Implications for Species Distribution Modelling." *Biological Reviews* 88, no. 1 (2013): 15–30. https://doi.org/10.1111/j.1469-185X.2012.00235.x.

Manuscript in submission

Wood, Simon N. *Generalized Additive Models: An Introduction with R*. 2nd ed. Chapman and Hall/CRC, 2017. https://doi.org/10.1201/9781315370279.

Wu, Zonghan, Shirui Pan, Fengwen Chen, Guodong Long, Chengqi Zhang, and Philip S. Yu. "A Comprehensive Survey on Graph Neural Networks." *IEEE Transactions on Neural Networks and Learning Systems* 32, no. 1 (2021): 4–24. https://doi.org/10.1109/TNNLS.2020.2978386.

Mauscript in submission